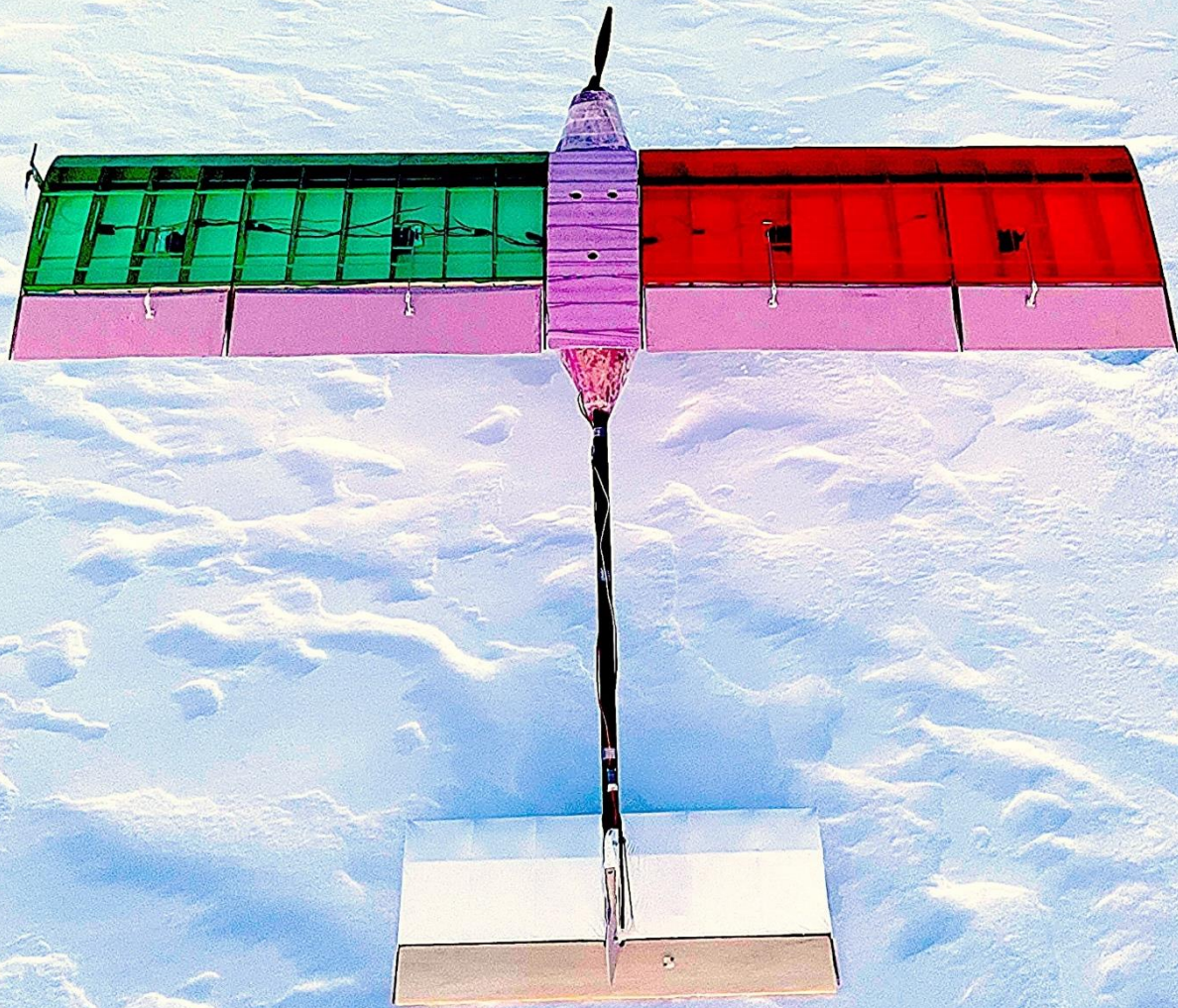




**WASHINGTON UNIVERSITY
IN ST. LOUIS**

“Ursa Major”



**AIAA DBF 2024-2025
DESIGN REPORT**



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Nomenclature

Abbreviations

AIAA	American Institute of Aeronautics and Astronautics	VLM	Vortex lattice method
AoA	Angle of attack	VT	Vertical tail
AR	Aspect ratio	WU	Washington University
AUW	All-up weight	WUDBF	Washington University in St. Louis Design/Build/Fly
CA	Cyanoacrylate	XPS	Extruded Polystyrene
CAD	Computer-aided design	Symbols	
CBDM	Component drag build-up method	α	Angle of attack (deg)
CF	Carbon fiber	λ	Taper ratio
CFD	Computational Fluid Dynamics	δ	Aileron deflection (deg)
CG	Center of gravity	ν	Kinematic viscosity (m^2/s)
DBF	Design/Build/Fly	ρ	Density (kg/m^3)
ESC	Electronic Speed Controller	A	Characteristic reference area (m^2)
FEA	Finite Element Analysis	b	Span (m)
GM	Ground Mission	c	Chord (m)
HQRS	Cooper-Harper handling qualities rating scale	C_D	Drag coefficient
HT	Horizontal tail	C_L	Lift Coefficient
Km_sim	Kinematic mission simulator	F	Drag force (kgf)
LiPo	Lithium Polymer	I_T	Optimal tail arm length
M1	Mission 1	MAC	Mean aerodynamic chord (m)
M2	Mission 2	Re	Reynolds number
M3	Mission 3	S	Wing area (m^2)
NACA	National Advisory Committee for Aeronautics	T	Thrust (kgf)
NiCad	Nickel Cadmium	T/W	Thrust-to-weight ratio
NiMH	Nickel-metal hydride	u	Velocity (m/s)
NP	Neutral Point	$\mathbf{V}$	Volume (m^3)
RC	Radio-controlled	V_∞	Freestream velocity (m/s)
RPM	Revolutions per minute	v_{max}	Max cruise velocity (m/s)
SM	Static margin	W	Weight (kgf)
TOFL	Takeoff field length	W/S	Wing loading (kg/m^2)
T-O	Takeoff	X	x-location (m)

1 Executive Summary

This report details the design, manufacturing, and testing of *Ursa Major*, the Washington University in Saint Louis aircraft for the 2024-25 American Institute of Aeronautics (AIAA) Design/Build/Fly (DBF) competition. This radio-controlled aircraft was designed to perform four X-1 Supersonic Test Flight Program missions. Mission 1 (M1) is a delivery flight that verifies the design and flight capabilities of the aircraft. Mission 2 (M2) is a captive carry flight of externally mounted fuel tanks and the X-1 test vehicle. Mission 3 (M3) is a launch flight demonstrating the successful release of the X-1 test vehicle into the designated landing area. Finally, the Ground Mission (GM) demonstrates all features of the X-1 flight test vehicle program through efficient installation of fuel tanks and test vehicles to flight vehicle release.

Ursa Major has been designed to meet all mission requirements while maintaining a high level of reliability and safety. The aircraft consists of a conventional, single motor, tricycle, high-wing aircraft. These requirements were chosen in response to the stated requirements. The tricycle configuration is selected due to the lack of takeoff requirement and allows for easier installation of the X-1 test vehicle. The high-wing configuration allows for greater stability during M2 and M3. A sensitivity analysis performed during the conceptual design phase indicated that an M2 payload of 2.5 kg spread over 2 external pylons on each wing and a M3 X-1 test vehicle maximum weight of 0.2 kg was optimal given size constraints and WUDBF's desire for a stable and low-risk design.

During the preliminary design phase, the aircraft was further refined via detailed aerodynamic and propulsion design. Lift, drag, and stability analyses were performed to size the aircraft and set the outer geometry. The wingspan was set at 1.83 m with a wing loading of around 8.78 kg/m² for M2 and 5.83 kg/m² for M3. Fuel tank and test vehicle placement were also determined such that the center of gravity shifts are minimized following deployment. A propulsion system was then selected that gives the aircraft a 1.18 thrust-to-weight ratio and predicted maximum cruise velocity of 31.50 m/s. This value varies through missions and is dependent on percentage of throttle used and payload on board.

During the detail design phase, the X-1 test vehicle, the vehicle release mechanism, and the pylon and fuel tank attachment systems were designed. The glider is locked in place using a servo-lock system that can be released and a pull switch to activate strobe lights. The fuel tanks are attached directly to the carbon fiber wing spar with two on each side at 0.7 kgs per bottle for a total M2 payload of 2.5 kg.

Based on subsequent performance simulations and test flight data, the aircraft was able to take off in 12.7 m for M2 using full throttle and 10.8 m for M3 without full throttle, around 80 percent. The aircraft was predicted to complete M2 in 1.92 minutes, well within the 5 minute time limit and yielding a predicted normalized score of 2.0 points when compared to other configurations WUDBF simulated. For M3, the aircraft was predicted to fly for a total of 4.61 minutes while completing 5 scoring laps, not including the lap after test vehicle release, yielding a predicted normalized score of 2.28 including the 2.5 point bonus. Endurance estimates aligned well with both the M2 and M3 flight times. Adding on the predicted M1 and GM scores, *Ursa Major* is estimated to score competitively with a total mission score of 5.78 points out of a possible 7. Based on previous years' performance, GM is predicted to earn 0.5 points normalized. Thorough ground and flight testing confirmed that the aircraft could meet all predicted performance goals.

2 Management Summary

2.1 Organization

WUDBF is a student run club with a faculty advisor. The executive board is split into both administrative and technical functions to ensure efficient workflow and operations. The roles and skills of each sub-team are further explained in Table 2.1. The organization of the sub-teams is depicted in Fig. 2.1.

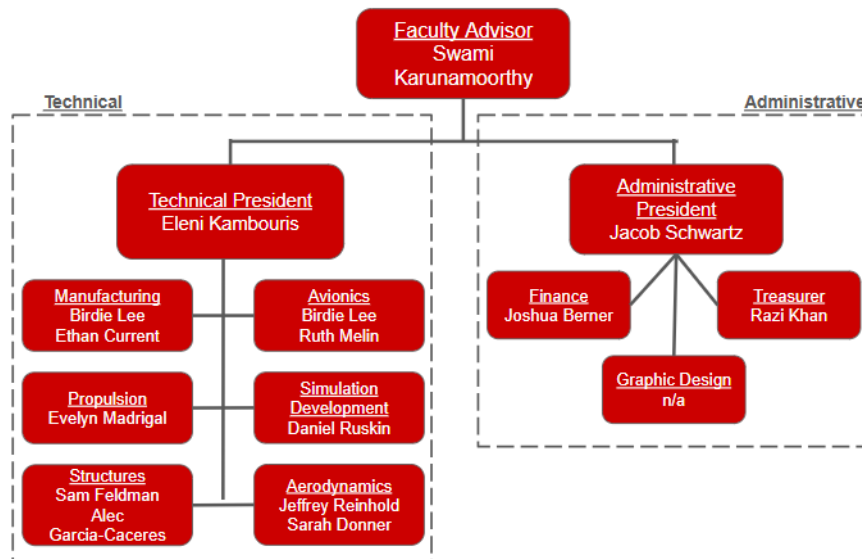


Figure 2.1: WUDBF Organizational Chart

Administrative responsibilities such as acquiring sponsorships, managing the budget, and graphic design for club marketing are overseen by the Administrative President. The Administrative President is also responsible for scheduling and leading general body meetings. The Administrative President manages administrative division, which includes three sub-teams: finance, graphic design, and treasury. The Technical President is responsible for approving technical decisions and oversees the technical division. The technical division includes six sub-teams: simulation and development, aerodynamics, structures, propulsion, manufacturing, and avionics. Each sub-team is managed by a sub-team lead who organizes meeting tasks and ensures their sub-team is on track to complete their components of the overarching design and testing process in a timely manner.

Table 2.1: Team roles and skills.

	Aerodynamics	Structures	Manufacturing	Avionics	Propulsion	Simulation Development
Roles	Design of the exterior geometry of the aircraft, as well as optimization of aircraft stability.	Design and modeling of the aircraft structure and any additional mission specific components.	Assembly of the aircraft and components in addition to materials testing.	Installation of electrical components, programming of flight controllers, and test flight data collection.	Design of the propulsion package and motor testing.	Development of simulation tools used for optimization and sensitivity analysis.
Skills	Knowledge of aircraft performance, stability and control, CFD (Ansys), and XFLR5.	Knowledge in CAD, structural FEA, and mass properties analysis.	Knowledge in laser cutting, fiberglass layups, hot-wire foam cutting, 3D printing, and balsa aircraft construction.	Knowledge in electrical design, soldering, Real Flight 8, data collection, and autonomous flight programming.	Knowledge in propulsion theory and analysis, Ecalc, and Remote Control (RC) Benchmark.	Knowledge in MAT-LAB/Simulink and Python.

2.2 Schedule

The team's schedule is depicted below by the Gantt chart in Fig. 2.2. WUDBF adheres to the chart to meet deadlines for the design and testing process. The executive board and leadership team meet weekly to ensure effective cross-functional communication.

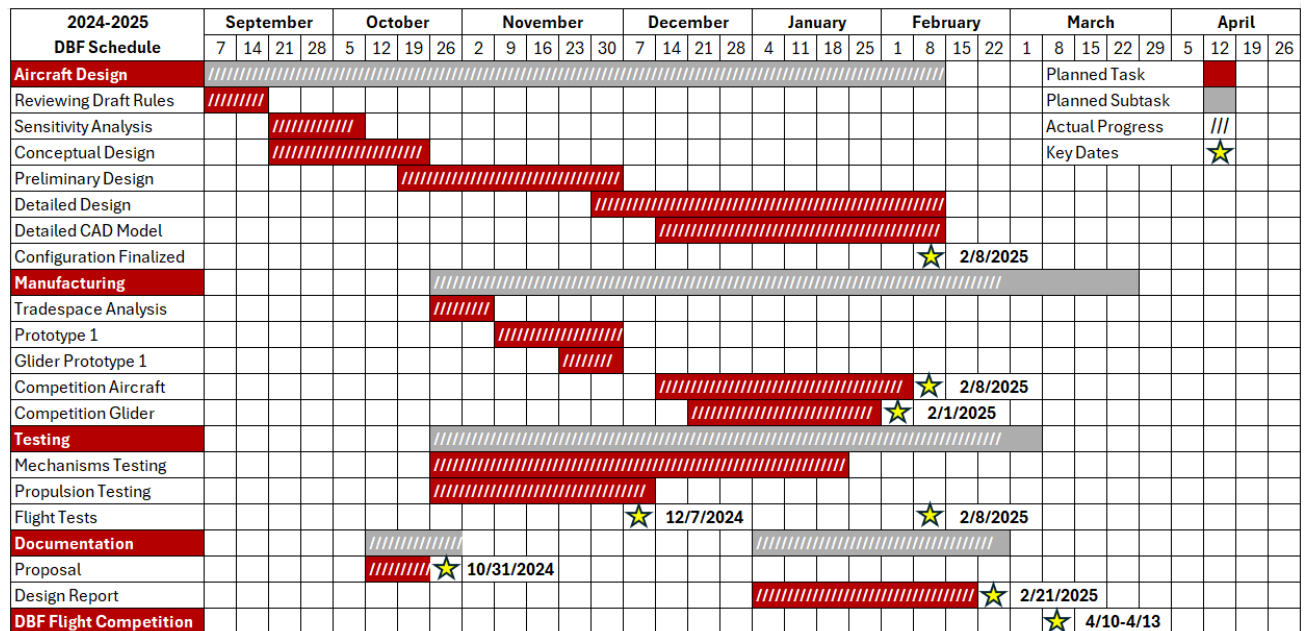


Figure 2.2: 2024-2025 Gantt Chart

3 Conceptual Design

For the 2024-25 Design/Build/Fly (DBF) competition, the AIAA has set the objective to design and build an aircraft for an X-1 Supersonic Flight Test Program. The aircraft must be capable of carrying fuel tanks and an X-1 test vehicle, while being able to launch an X-1 test vehicle capable of autonomous flight. Upon receipt of the competition rules [1], WUDBF began the conceptual design phase starting with identification of key mission requirements and translation of those mission requirements into design requirements, followed by a scoring

sensitivity analysis and configuration selection via a series of decision matrices.

3.1 Mission Requirements

The 2024 competition consists of four missions; one ground mission and three flight missions. The same course is used for all flight missions and is shown in Fig. 3.1. Both the flight missions and the ground mission follow the same setup procedure. The aircraft must enter the staging area in its parking configuration, then be deployed into its flight configuration and loaded with the mission payloads within five minutes.

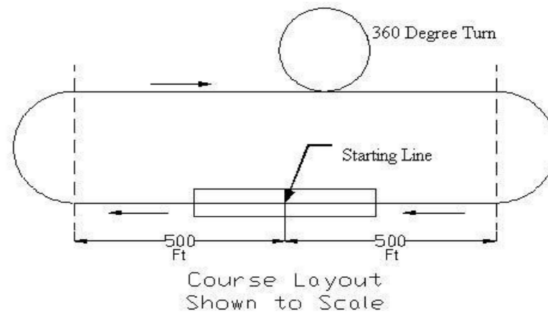


Figure 3.1: Competition course layout.

3.1.1 Mission 1 - Delivery Flight

There is no payload for Mission 1. The aircraft must complete three laps of the course within a five-minute window. Landing may occur outside the window, but the aircraft must successfully land to get a score. Mission 1 scoring is calculated as follows:

$$M1 = 1.0 \quad (3.1)$$

The maximum score for mission one is **1.0 point** awarded for completion.

3.1.2 Mission 2 - Captive Carry Flight

The payload for the Captive Carry Flight consists of the X-1 test vehicle and the fuel tanks. The aircraft must complete three timed laps of the course within a five-minute window. A lap is counted when the aircraft crosses the start/finish line. Landing may occur outside the window, but the aircraft must successfully land to get a score. Mission 2 scoring is calculated as follows:

$$M2 = 1.0 + \frac{(fuel_weight/time)_{WUDBF}}{(fuel_weight/time)_{max}} \quad (3.2)$$

Where *fuel_weight* is the weight of the fuel tanks, and *time* is the time it takes to complete three laps. The WUDBF and max subscripts denote the raw scores achieved by WUDBF and the team that scored highest on M2, respectively. A total of **2.0 points** are possible on M2, with one coming from completion and a second from performance.

3.1.3 Mission 3 - Launch Flight

The payload for the Launch Flight consists of the X-1 test vehicle and the fuel tanks. After finishing the first lap or after any subsequent lap, the aircraft must launch the X-1 test vehicle after crossing the start/finish line, before the upwind turn, and at an altitude between 200-400 ft. The aircraft must complete its full lap within the five-minute time window after the launch. A lap is counted when the aircraft crosses the start/finish line. Landing may occur outside the window, but the aircraft must successfully land to get a score.

For bonus points to be achieved in Mission 3, the X-1 test vehicle must reach stable flight after launch, perform a

180-degree turn, and fly a specific orbit or descending pattern. Further, the X-1 test vehicle must have lights that come on after the vehicle leaves the aircraft. Bonus points are allocated based on the X-1 test vehicle coming to rest within the five-minute time window in one of the bonus boxes detailed in Fig. 3.2. 2.5 bonus points are given for coming to rest in the '2.5 points' box, 1 bonus point is given for coming to rest in the '1.0 point' box, and no bonus points are given otherwise. If the vehicle comes to rest outside of the safety line, then the total Mission 3 score is 0.

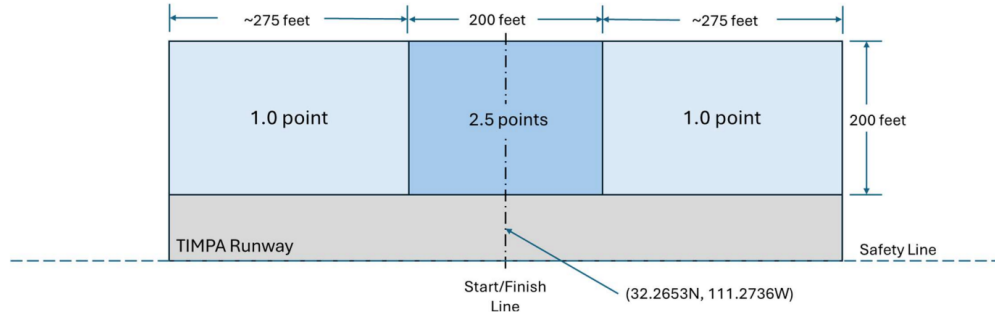


Figure 3.2: X-1 test vehicle bonus box layout.

Mission 3 scoring is calculated as follows:

$$M3 = 2.0 + \frac{(\#laps + bonus_box / X1_weight)_{WUDBF}}{(\#laps + bonus_box / X1_weight)_{max}} \quad (3.3)$$

Where $\#laps$ is the number of laps completed before the X-1 test vehicle is launched, $bonus_box$ is the bonus box score of 2.5, 1, or 0, and $X1_weight$ is the X-1 test vehicle weight. The WUDBF and max subscripts denote the raw scores achieved by WUDBF and the team that scored highest on M3, respectively. The maximum score on Mission 3 is **3.0 points**, with two points awarded for completion and a third for performance.

3.1.4 Ground Mission - X-1 Flight Test Program Demonstration

The Ground Mission serves as a timed ground demonstration of the aircraft's Mission 2 and 3 capabilities. The Ground Mission begins with the aircraft on its landing gear without the external fuel tank pylons, fuel tanks, and X-1 test vehicle installed. The aircraft may have blocks beneath it to allow for easy underneath access. The mission is divided into three phases. In the first, the mission official will say "GO" and begin the timer. The ground crew member will install all external fuel tank pylons. The official will then pause the timer and verify that the number of pylons installed is the same as the number of pylons declared in Tech Inspection. In the second phase, the official will again say "GO" and the ground crew member will install the fuel tanks and the X-1 test vehicle, then close and secure all hatches. The timer for the mission will stop, and the aircraft will be checked to ensure that the correct number of fuel tanks are installed, the fuel tanks and the X-1 test vehicle are installed properly, and the hatches are closed. The pilot will then verify the flight controls. In the third phase, the pilot will release the X-1 test vehicle from the aircraft, with underneath padding previously added by the ground crew member if necessary. The total mission time will be recorded, as long as the lights properly turn on after the release of the X-1 test vehicle. Scoring is as follows:

$$GM = \frac{(time)_{min}}{(time)_{WUDBF}} \quad (3.4)$$

Where *time* is the total Ground Mission Time. The WUDBF and min subscripts denote the raw scores achieved by WUDBF and the team with the lowest GM time, respectively. The maximum achievable score for the Ground Mission is **1.0 points** based on performance.

3.1.5 Total Competition Score

Mission 1, 2, 3, and the Ground Mission scores are combined to give the total mission score:

$$\text{Total Mission Score} = M1 + M2 + M3 + GM \quad (3.5)$$

The maximum achievable Total Mission Score is **7.0 points**, with four awarded for completion and three for performance. The scores from the Proposal and Design Report are combined to produce the total report score:

$$\text{Total Report Score} = 0.15 \times \text{Proposal Score} + 0.85 \times \text{Design Report Score} \quad (3.6)$$

The final Competition Score is calculated as follows:

$$\text{Competition Score} = \text{Total Mission Score} * \text{Total Report Score} + P \quad (3.7)$$

Where *P* is the participation score, with one point awarded for attending the fly-off, a second for completing tech inspection, and a third for attempting a flight mission.

3.1.6 General Mission Requirements

In addition to the requirements specific to each mission, there are also a number of other rules of note that apply to the competition aircraft during all missions:

- The aircraft's wingspan cannot exceed six feet in the flight configuration.
- At least two external fuel tanks must be attached to the aircraft during M2 and M3, and the fuel tanks must be commercially available bottles with a capacity of at least 16 fl oz.
- Removable pylons must be used to securely attach external fuel tanks during all phases of flight.
- The X-1 test vehicle must have a maximum weight of 0.55 pounds.
- The X-1 test vehicle must be attached underneath the aircraft fuselage between left and right external fuel tanks. A bomb bay opening may be used in the lower fuselage to allow the X-1 test vehicle to launch, but the opening must be closed with a cover during M1 and during the beginning of GM. The cover must be permanently attached to the aircraft.
- The aircraft may utilize either Nickel Cadmium (NiCad)/Nickel Metal Hydride (NiMH) OR Lithium Polymer (LiPo) batteries for propulsion, but the propulsion power total stored energy cannot exceed 100 watt-hours.
- The flight missions must be flown in order; a new mission cannot be flown until the team has obtained a successful score for the preceding mission.

3.2 Translation to Design Requirements

Based on the mission requirements outlined above, WUDBF formulated design requirements for each mission. These design requirements are broken down in Table 3.1.

Table 3.1: Translation of mission requirements into design requirements.

Mission	Mission Requirement	Design Requirement
M2	Minimize lap time	High cruise velocity, high T/W, low drag
	Maximize fuel weight	Large wing area, sufficient internal volume
	Carry X-1 test vehicle	Large wing area
M3	Maximize laps	High cruise velocity, high T/W, low drag, high propulsive efficiency, high battery capacity
	Carry X-1 test vehicle	Large wing area
	Minimize X-1 test vehicle weight	Simple X-1 test vehicle design
	Land X-1 test vehicle in highest scoring bonus box	Stable autonomous flight
GM	Minimize loading and launch time	Optimized payload storage configuration, easy access to internal storage, easy switching between configurations, easy launch of X-1 test vehicle
General	Complete every mission	Low risk design, high stability
	Max propulsion energy of 100 watt-hours	High efficiency, constrains maximum propulsion power
	Max wingspan of 6 ft	Constrains wingspan

3.2.1 Analysis of Design Requirements

Mission 2: Higher scores in M2 are obtained by maximizing fuel weight while minimizing lap time. This leads to contradicting design considerations. To increase the fuel weight, the aircraft must increase its wing area to support a realistic wing loading. Increasing the wing area leads to more drag, causing a decrease in cruise velocity and thus hurting lap time. Therefore, a balance must be struck between maximizing fuel weight and maximizing cruise velocity.

Mission 3: Higher scores in M3 are obtained by maximizing laps, minimizing the X-1 test vehicle weight, and maximizing the bonus box score from landing the X-1 test vehicle. Maximizing laps requires higher cruise velocity, and higher cruise velocity occurs with a smaller wing area. Fortunately, the mission requirement of minimizing the X-1 test vehicle weight does not require an increase in wing area. However, the X-1 test vehicle must also be stable enough for autonomous flight, which may mean that more internal components are needed, increasing the weight of the X-1 test vehicle. Thus, there is a trade-off between minimizing X-1 test vehicle weight and ensuring that the X-1 test vehicle is stable enough to launch and land in a bonus box. Further, the aircraft must also be able to carry an X-1 test vehicle capable of autonomous flight, which means that the wing area should be larger. Therefore, similar to in Mission 2, the design must also balance between maximizing cruise velocity and maximizing wing area.

Ground Mission: Minimizing GM time requires the aircraft to have hatches that facilitate easy installation of pylons, fuel tanks, and the X-1 test vehicle. Given that a successful launch of the X-1 test vehicle is required in both GM and M3, it is crucial for the designs of the X-1 test vehicle and the aircraft to facilitate an easy launch.

Minimal Risk: Though mission performance is an important part of the competition score, over half of the mission points are awarded for completion. As such, the most important aspect of the aircraft's design is its overall stability and reliability.

Constraints: The 100 Wh propulsion energy limit imposes a restriction on the power of the propulsion system, meaning that high efficiency is a must. The wingspan is constrained by the 6 ft wingspan limit, which makes it more difficult to carry the X-1 test vehicle.

3.3 Scoring Model and Sensitivity Analysis

Considering the design requirements above, WUDBF determined that no obvious solution presented itself to the design problem. Maximizing cruise velocity to decrease lap times in M2 and increase the number of laps in M3 would lead to a smaller, faster aircraft. However, maximizing payload in M2 and increasing the wing area to carry an X-1 test vehicle capable of autonomous flight in M3 would generate a larger, slower aircraft. And while minimizing the weight of the X-1 test vehicle was a design consideration for M3, it was also essential that the X-1 test vehicle had the internal components needed to safely launch and land in the proper box. Given these conflicting strategies, WUDBF designed a scoring model and conducted a sensitivity analysis to determine which factors would contribute most to the total score.

3.3.1 Scoring Model

To create the scoring model, WUDBF swept through different values of four independent variables: fuel weight in M2, X-1 test vehicle weight in M3, maximum cruise velocity in M2 and M3, and wing area. These variables were chosen because they could be utilized, along with other assumed values, to estimate every mission score. WUDBF calculated four-dimensional matrices for each mission, containing a score for each possible combination of values. Sweeping through these four variables also allowed for calculations of all mission constraints and self-prescribed constraints, which included minimums/maximums for takeoff length, wing loading, power, and endurance. These constraints were applied to the mission score matrices, eliminating all matrix positions corresponding to combinations of values that did not satisfy every constraint. Justifications for all assumed values and self-prescribed constraints are provided in Table 3.2.

Table 3.2: Table of key assumptions and justifications for the scoring model.

Assumption	Justification
Drag Coefficient ($C_{D,min}$) is 0.08.	Based on previous WUDBF aircraft testing and preliminary CFD analysis.
C_L is 0.96.	From pilot interviews, a Lift/Drag ratio of 12 is typical for RC aircraft.
Wing area varies between 0.1 and 1 m ² .	An aircraft with a wing area of 0.1 m ² is unlikely to take off with the required payload in M2, and an aircraft with a wing area of 1 m ² is unlikely to satisfy the 6 ft wingspan limit.
The maximum wing loading is 10 kg/m ² .	Test pilot feedback on previous aircraft suggests a wing loading heavier than 10 kg/m ² will make the aircraft very difficult to control.
The maximum power is 2000 W.	A 2000 W motor will deplete a 100 Wh battery in 3 minutes, and therefore will likely not score well on the endurance mission, if it is able to complete 3 laps at all.
The minimum X-1 test vehicle weight is 0.05 kg.	It is unlikely that a test vehicle with a weight below 0.05 kg could achieve stable autonomous flight.
The maximum cruise velocity is 45 m/s.	Test pilot feedback on previous aircraft suggests a cruise velocity greater than 45 m/s will make the aircraft very difficult to control.
The maximum takeoff distance is 30 m.	30 m is well above the takeoff distances achieved by previous WUDBF aircraft, and any takeoff distance above 30 m means that the aircraft wastes too much time on the ground.
Minimum aspect ratio is 4.	It was determined based on previous WUDBF aircraft testing that an aspect ratio under 4 produces an infeasible aircraft design.
Battery capacity is 100 Wh.	Previous WUDBF aircraft have achieved the best performance while using the maximum battery capacity possible.
Wing span is 1.83 m.	WUDBF determined that the maximum wing span of 1.83 m (6 ft) would be needed to ensure stable flight while carrying the fuel tanks and X-1 test vehicle.
Lap length is approximately 762 m.	Based on the course diagram.

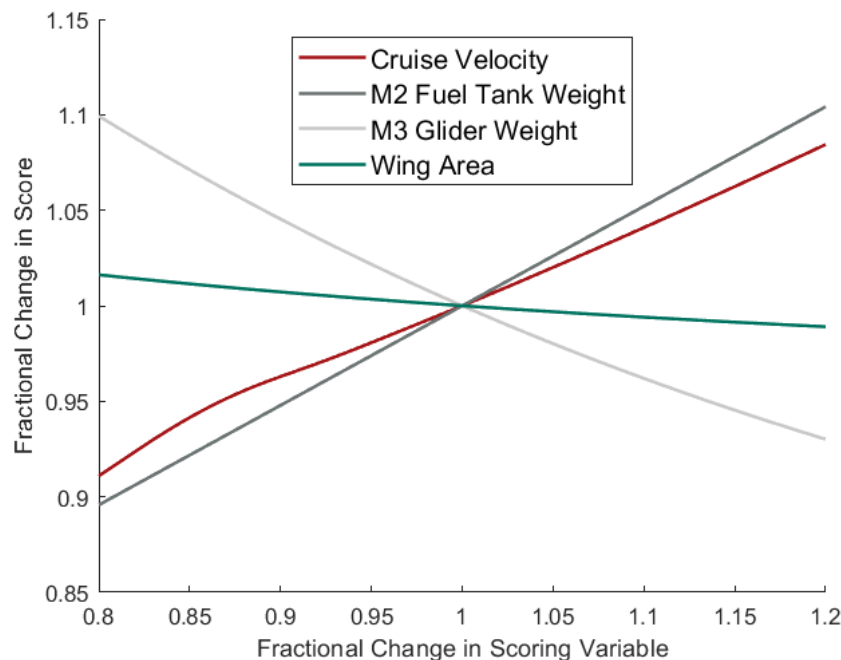
After constraints were applied, the mission score matrices were normalized by the highest score in each mission to simulate the competition rules. Then, the score matrices were summed to create a total score matrix. The four input values corresponding to the maximum total score were used to generate initial design targets, displayed in Table 3.3. One can see that the maximum possible score was achieved for M2, while the score for M3 depended greatly on the bonus box score.

Table 3.3: Initial design targets.

Design Specifications	Initial Target Value
Wing Area	0.81 m ²
Power	1842 W
M2 Fuel Tank Weight	2.50 kg
M2 All-up Weight (AUW)	8.06 kg
M2 Takeoff Distance	5.14 m
M2 Wing Loading	9.92 kg/m ²
M2/M3 Max Cruise Velocity	31.3 m/s
M2 Normalized Score	2
M3 X-1 Test Vehicle Weight	0.05 kg
M3 Number of Laps Before X-1 Launch	7
M3 Normalized Score	2.11 (0 bonus), 2.43 (1 bonus), 2.92 (2.5 bonus)

3.3.2 Sensitivity Analysis

The design targets were used to guide the initial conceptual design and were further refined in the preliminary design phase. Using the optimal point of variables corresponding to the maximum total score, a sensitivity analysis was done to see how deviations in each parameter would alter the total score. Understanding how slight alterations in inputs affect the total score would help guide future design decisions. Results of the analysis are shown in Fig. 3.3.

**Figure 3.3: Sensitivity analysis results.**

The sensitivity analysis confirmed the importance of maximizing cruise velocity, maximizing the fuel tank weight in M2, and minimizing the X-1 test vehicle weight in M3. However, as explained in Section 3.2.1, it is not feasible to maximize cruise velocity without decreasing fuel tank weight, and vice versa. After considering these contrasting variables, WUDBF decided to emphasize a high wing area, and thus a high M2 fuel tank weight, at the cost of decreased velocity. This decision was based on the fact that M2 fuel tank weight was slightly more important than cruise velocity in the analysis, as indicated by the slightly higher slope of the M2 fuel tank weight line. Additionally, the sensitivity analysis indicated that significant emphasis should be placed on minimizing the M3 X-1 test vehicle weight, given its large slope in the analysis and minimal contrasting objectives. At the same time, the M3 glider weight must not be minimized such that it hurts the stability of the X-1 test vehicle and its ability to land in a bonus box. In the end, WUDBF prioritized building a larger, slower aircraft instead of a smaller, faster aircraft, while emphasizing a light but stable X-1 test vehicle in M3.

3.4 Configuration Selection

Once initial design targets were determined, decision matrices were then used to select the overall aircraft configuration as well as the configuration of major subsystems. All decision matrix weights were determined based on design goals and competition requirements. Individual criterion scores were then determined based on team knowledge and experience from previous competitions. The individual criterion scores were multiplied by the weights and summed together to determine the total weighted score for each configuration. A selection was then made by comparing the total weighted score.

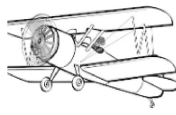
3.4.1 Configuration Selection

WUDBF first considered three overall aircraft configurations including the monoplane, biplane, and blended body configurations. Key design criteria included aerodynamic efficiency, stability and control, wing loading, and manufacturability. The highest weights were assigned to flight speed and stability and control as speed is important to earning points, and stability is a main driver of overall aircraft design. Using these criteria, the following configurations were compared in Table 3.4.

- **Monoplane:** Provides good aerodynamic efficiency, stability and control, and is likely the most feasible configuration. However, available wing area within the span limitation is the lowest of the considered configurations.
- **Biplane:** Potentially allows for more lift within the span limitation by maximizing available wing area. However, the increased drag and added structural weight associated with the configuration decreases efficiency.
- **Blended Body:** Allows for increased wing area while still maintaining relatively good aerodynamic efficiency. However, the complex geometry of the blended body reduces design and manufacturing feasibility. Additionally, the blended body configuration would have limitations in regard to stability and control.

Translating these judgments to the decision matrix resulted in Table 3.4 below:

Table 3.4: Selection matrix primary configuration.

Selection Criterion	Weight (%)	Score		
		 Monoplane	 Biplane	 Blended Body
Flight Speed	25	3	1	5
Drag	15	4	1	5
Wing Loading	15	4	5	3
Stability and Control	25	5	2	5
Manufacturability	20	5	4	1
Weighted Score		4.2	2.5	3.9
Rank		1	3	2

Ultimately, the monoplane configuration was selected due to its superior efficiency over the biplane and easier manufacturability compared to the blended body.

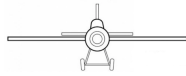
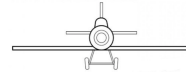
3.4.2 Wing Placement

WUDBF next considered wing placement by comparing high-, mid-, and low-wing configurations. Takeoff capability and ease of loading were weighted highest followed by stability, flight speed, and maneuverability.

- **High-Wing:** Possesses the best stability and control due to its tendency to self-correct. Also is the most feasible configuration since WUDBF used it successfully in the past. However, access to the fuselage from above is restricted, increasing payload loading complexity and GM time.
- **Mid-Wing:** Has stability and control characteristics between that of the low- and high-wing configurations. However, this configuration necessitates that the structural support for the wing runs through the center of the fuselage. This results in a decreased payload capacity and potentially increased payload loading complexity because the bar might conflict with either the battery or the payload mass.
- **Low-Wing:** Because the low-wing configuration allows for easy access from above, it has superior payload loading capability. However, it is the least stable configuration due to the CG being above the wing. An unstable configuration would require constant maneuver corrections, ultimately slowing down the aircraft.

The resulting decision matrix is shown in Table 3.5 below:

Table 3.5: Selection matrix wing placement.

Selection Criterion	Weight (%)	Score		
				
		High Wing	Mid Wing	Low Wing
Maneuverability	10	3	4	5
Stability and Control	40	5	4	2
Ease of Landing	5	3	4	5
Manufacturability	15	3	2	3
Flight Speed	30	3	4	5
	Weighted Score	4.1	3.7	3.2
	Rank	1	2	3

Ultimately the high wing was selected because it was the most stable orientation, and its manufacturing problems, such as fuselage attachment were deemed simple enough to overcome.

3.4.3 Empennage Configuration

WUDBF then selected the empennage configuration out of the conventional tail, the T-tail, the H-tail, and the V-tail. The main design criteria for empennage configuration were determined to be stability and control, flutter, drag, wake interference during takeoff, and feasibility. Using these criteria, the following configurations were compared in Table 3.6.

- **Conventional:** Simple to design and manufacture. Also possesses good stability and control and flutter characteristics.
- **T-tail:** Keeps the horizontal tail (HT) in undisturbed air, which would reduce the drag. However, the high-mounted HT can also generate low flutter speeds and asymmetric lift and yaw which induces high torsional loads. Additionally, having the HT at the tip of the vertical tail (VT) introduces additional complexity into the structure and manufacturing process.
- **H-tail:** Offers increased HT efficiency due to the endplates. However, the flutter speed of the H-tail is reduced due to the weight of the VTs being located at the tips of the HT. Additional control and structural complexity is introduced by having two VTs, decreasing feasibility.
- **V-tail:** Possesses good flutter characteristics and reduces the wake interference at takeoff. However, the V-tail control system is significantly harder to design and implement.

Table 3.6: Selection matrix empennage configuration.

Selection Criterion	Weight (%)	Score			
					
		Conventional	T-Tail	U-Tail	V-Tail
Drag	30	4	4	2	5
Stability and Control	40	3	4	5	4
Feasibility	30	5	3	2	2
	Weighted Score	3.9	3.7	3.4	3.7
	Rank	1	3	4	2

The conventional tail configuration was chosen to maximize manufacturability. Additionally, the conventional tail configuration is the design WUDBF has successfully employed in the past.

3.4.4 Landing Gear Configuration

The landing gear was then chosen based on ground maneuverability, drag, structural strength, feasibility, and takeoff capability. Using these criteria, the following configurations were compared in Table 3.7.

- **Tricycle:** Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.
- **Bow Tricycle:** Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.
- **Strut Tail-dragger:** Three struts or fairings independently connected to the fuselage. Provides good ground maneuverability and provides the option to allow the landing gear to retract into the fuselage. However, the structural strength of the front landing gear and takeoff capability of this configuration is a concern.
- **Bow Tail-dragger:** Two front wheels connected to the fuselage via a common strut or fairing and one rear wheel connected to the tail. Offers the same improved takeoff capability as the strut tail-dragger, while also providing better structural support. Ground maneuverability of this configuration is limited.

Table 3.7: Selection matrix landing gear configuration.

		Score			
					
Selection Criterion	Weight (%)	Tricycle	Bow Tricycle	Strut Tail-Dragger	Bow Tail-Dragger
Maneuverability	30	5	5	2	2
Stability and Control	25	4	4	4	3
Takeoff Length	25	2	3	3	5
Ease of Landing	10	4	5	3	2
Flight Speed	10	5	5	3	3
Weighted Score		3.9	4.25	2.95	3.1
Rank		2	1	4	3

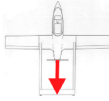
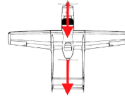
The Bow Tricycle configuration was chosen because it scored the highest by performing very well in ground maneuverability and feasibility compared to the other designs.

3.4.5 Propulsion Configuration

The propulsion system was chosen based on its efficiency, thrust, weight, component interference, and drag. The following propulsion configurations were considered and the constructed decision matrix is shown below in Table 3.8.

- **Tractor:** Single motor on the front of the aircraft. Provides the highest efficiency, lowest weight and drag, and is the most feasible. Ground clearance could be a concern depending on the required propeller size and height of front landing gear.
- **Pusher:** Single motor aft of the fuselage. Provides low weight and drag but has a decreased ground clearance (especially during rotation) and low efficiency due to fuselage interference.
- **Twin:** Two motors which are mounted on the wings. Provides improved ground clearance due to reduced propeller size and reasonable efficiency. Also offers torque balancing and is reasonably feasible. However, drag is increased resulting from the placement of the motors on the leading edge of the wing. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.
- **Push/Pull:** One motor at the front and aft of the fuselage. Offers low weight and drag. However, is the least efficient due to propwash interference, has the same ground clearance concerns as the tractor and pusher configurations, and is the least feasible. Weight is also a concern because of the additional electronics and structural strength needed to support the second motor.

Table 3.8: Selection matrix propulsion configuration.

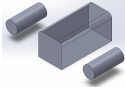
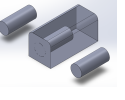
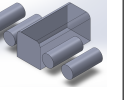
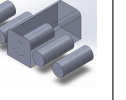
Selection Criterion	Weight (%)	Score			
					
		Tractor	Pusher	Twin	Push/Pull
Maneuverability	35	5	4	3	3
Stability and Control	25	3	3	5	4
Takeoff Length	15	5	5	3	3
Ease of Landing	10	3	3	5	1
Flight Speed	15	3	4	3	5
	Weighted Score	4	3.8	3.7	3.35
	Rank	1	2	3	4

The tractor configuration was ultimately chosen because it was thought to maximize efficiency while minimizing weight and drag. Additionally, the tractor configuration is the most feasible design and is one that WUDBF has successfully employed in the past.

3.4.6 Load Configuration

The load configuration was chosen based on its longitudinal weight distribution, wing structure, ease of loading, and fuselage space. The following loading configurations were considered and the constructed decision matrix is shown below in Table 3.9. The notation is in the form number of bottles on wings x number of bottles in the fuselage.

Table 3.9: Selection matrix loading configuration.

Selection Criterion	Weight (%)	Score			
					
		2x0	2x1	4x0	4x1
Longitudinal weight distribution	20	4	3	5	2
Wing Structure	20	4	3	5	2
Ease of Loading	35	4	3	5	2
Fuselage space	25	5	3	5	3
	Weighted Score	3.4	2.4	4.0	1.8
	Rank	2	3	1	4

The 4X0 configuration was ultimately chosen because it was thought to be easy to load, and provided empty space in the fuselage for electrical components.

3.4.7 Glider Configuration

WUDBF chose a near-traditional design for the glider. Traditional aspects of glider design like a high aspect ratio and low profile fuselage were selected to permit a shallow glide slope, allowing for maximum glide distance from the detachment point. The glider features a polyhedral wing to ensure high roll stability without the requisite

strength sacrifice near the wing-fuselage attachment point that results from a standard dihedral configuration. An untraditional inverted vertical stabilizer was chosen to allow for closer mounting to the plane without sacrificing yaw authority.

3.5 Final Conceptual Design

The final conceptual design consists of a high-wing, monoplane, bow-tricycle aircraft with a conventional tail, a propulsion system in the tractor configuration, and a payload configuration of 4 bottles under the wings. The aircraft is designed to carry a payload of 2.5 kg under the wings and a glider mass of 0.2 kg. The targeted max cruise velocity is 31.3 m/s.

4 Preliminary Design

During the preliminary design phase, the initial aircraft concept and configuration from the conceptual design phase were further developed. Key aspects of the design including the overall sizing, wing geometry, airfoil, and propulsion package were refined following several trade studies.

4.1 Design Methodology

The full design methodology employed by WUDBF is outlined in Fig. 4.1 below.

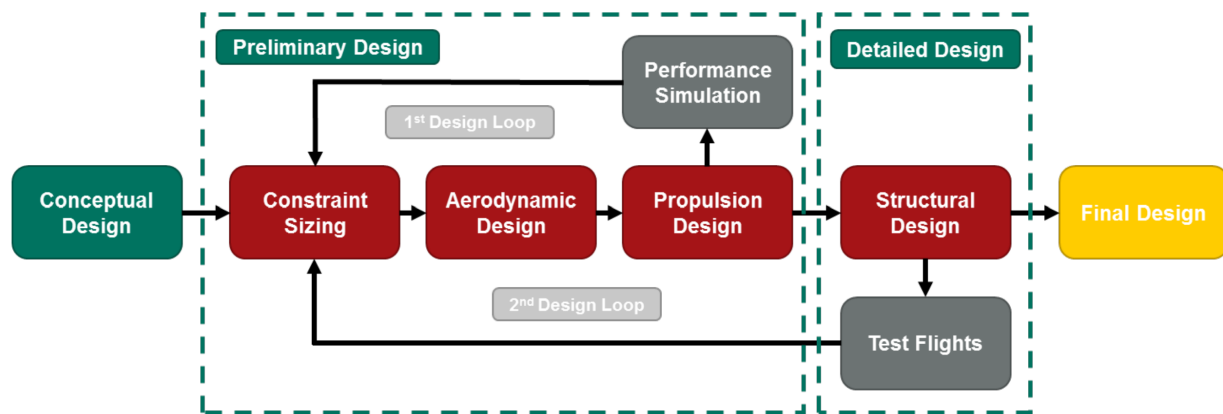


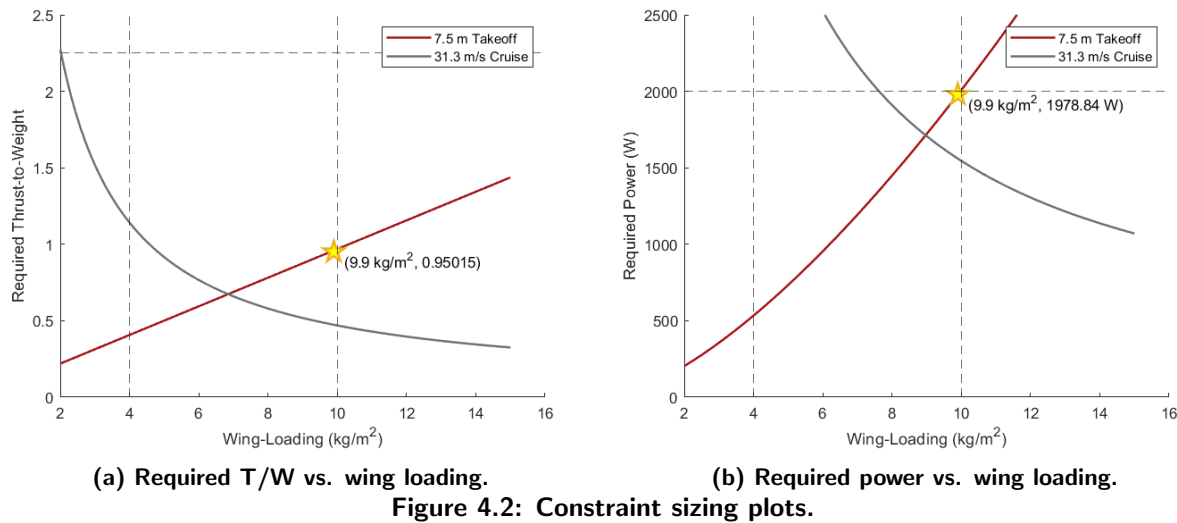
Figure 4.1: WUDBF design methodology.

4.2 Constraint Sizing

An initial constraint sizing was performed by plotting the required thrust-to-weight ratio to achieve a 7.5 m takeoff distance and 31.3 m/s cruise speed in M2, the mission where the aircraft has the heaviest AUW, for a range of wing loadings using the equations developed in Gudmundsson [2]. These plots are shown in Fig. 4.2. The takeoff and cruise speed targets were based on the initial design targets determined by the scoring analysis in Section 3.3.1. Though a takeoff distance of 5.14 m was calculated as a design target, a takeoff distance of 7.5 m was used for the constraint sizing. This is because WUDBF has often exceeded predicted takeoff distance targets. Further, the lack of an official takeoff constraint in this year's competition will lead to the pilot placing less emphasis on achieving a low takeoff distance.

Since the aircraft must be propeller driven using an electric motor, it is useful to convert T/W to required electrical power using initial estimates of weight, mechanical efficiency, and propulsive efficiency. This meant that the upper power constraint of 2000 W listed in Table 3.2 could be applied in the constraint sizing, along with an upper T/W constraint of 2.25.

In the leftmost plot, the upper T/W constraint is obeyed by all wing loading values. However, in the rightmost



plot, the upper power constraint is only obeyed by wing loading values from around 7.5 kg/m² to the upper constraint of 10 kg/m².

Ultimately, a wing loading of 9.9 kg/m² was chosen for the design point. This aligns with the initial design target wing-loading of 9.92 kg/m² from the scoring analysis in Section 3.3.1. Additionally, in previous competitions, the final WUDBF aircraft has consistently exceeded the desired weight. Thus, designing an aircraft with a slightly higher wing loading allows for a larger structural weight margin (given that propulsion and payload weight are essentially constant).

From the chosen wing-loading of 9.9 kg/m², the minimum required power was set at about 1979 W and the minimum required T/W ratio at about 0.950 (as indicated by the stars in the figures above). Note that from the plots, the required T/W and power values are determined by the takeoff; the required power at cruise is only about 1560 W. Thus, the propulsion system could be designed to cruise at a lower throttle setting to increase endurance.

4.3 Initial Weight Estimation

The AUW of the aircraft was estimated using a breakdown of individual components. The weight of the wing, HT, VT, and fuselage were estimated using previous aircraft, which were built with similar methods and materials. Battery, motor, servo, electronics, central carbon fiber (CF) rod, landing gear, and pylon weights were determined by weighing representative components already possessed by WUDBF. The M2 fuel tank weight was determined using the optimization program described in Section 3.3.1, while the M3 fuel tank weight was estimated based on the weight of empty 16 fl oz fuel tanks. The X-1 test vehicle weight was determined by weighing the estimated necessary manufacturing and avionics components.

Table 4.1: Component weight breakdown.

Component	Weight [kg]
Propulsion Battery	0.643
Motor	0.424
Fuselage	1.285
Electronics	0.150
Avionics Battery	0.108
Wing Left	0.631
Wing Right	0.631
Rear Landing	0.090
Electronic Steering Control	0.080
Front Landing	0.040
M2 Pylons	0.352
M2 Fuel Tanks	2.5
M2 X-1 Test Vehicle	0.2
M3 Pylons	0.352
M3 Fuel Tanks	0.1
M3 X-1 Test Vehicle	0.2
Total M1	4.082
Total M2	7.134
Total M3	4.734

Based on this analysis, the predicted AUW of the fully loaded aircraft during M1 is 4.082 kg, during M2 it is 7.134 kg, and during M3 it is 4.734 kg.

4.4 Aerodynamic Performance

The aerodynamic design of the aircraft was iteratively improved to ensure that it is stable, controllable, and achieves maximum aerodynamic performance. XFLR5 [3], MATLAB [4], and Ansys [5] were the primary tools used for aircraft aerodynamic analysis. The size and shape of the main wing and stabilizers were driven by iterative static and dynamic stability analyses. Control surfaces and high lift devices were dimensioned to improve handling characteristics and minimize takeoff distance, respectively.

4.4.1 Main Airfoil Selection

In order to select the main airfoil, the predicted Reynolds number (Re) of the aircraft was first calculated using Eq. 4.1 [2].

$$Re = \frac{c \cdot V_{\infty}}{\nu} \quad (4.1)$$

Where c is the mean aerodynamic chord [m], V_{∞} is the freestream velocity [m/s], and ν is the kinematic viscosity of air [m^2/s]. Kinematic viscosity was calculated from the average temperature in April (27 °C) and altitude (807 m) in Tucson which yielded a kinematic viscosity of $1.729 \cdot 10^{-5} m^2/s$. These conditions yielded a Reynold's number of 1,120,000.

Airfoil selection was then performed by selecting several different airfoils with maximum C_L/C_D near a Reynold's number of 1,120,000. Additionally, the search was restricted to airfoils with a minimum thickness of 10% and

a maximum camber of 8% to ensure that the airfoil could accommodate structural and system integration and be reasonably manufactured with the processes available. Five promising airfoils were selected for comparison as shown in Table 4.2. Once these potential airfoils were identified, analysis of each airfoil was performed in XFOil

Table 4.2: Airfoil selection options [6].

Airfoil	Max CL/CD	Max. Thickness [% of chord]	Max. Camber [% of chord]
Clark X	102.7	11.7	3.3
Clark Y	98.7	11.7	3.4
NACA 1412	73.5	12.0	1.0
NACA 2412	87.3	12.0	2.0
USA-35B	106.0	11.6	4.1

[7] and XFLR5 to determine which provided a favorable lift coefficient, lift-to-drag ratio, and drag bucket at a Reynold's number of 510,000.

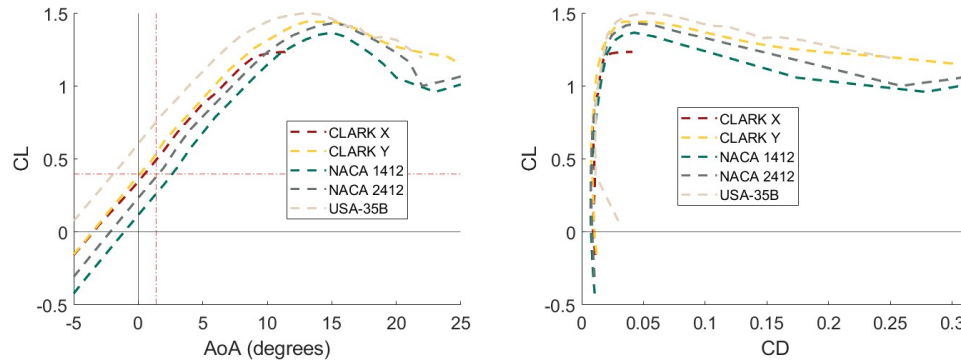


Figure 4.3: Airfoil C_L vs. AoA (left) and C_L vs. C_D (right).

As shown in Fig. 4.3, the USA-35B possessed the highest lift coefficients and widest drag bucket. While the lift generated from this airfoil would improve takeoff (T-O) performance, it would also produce excessive lift at the target cruise velocity of 31.3 m/s, meaning the aircraft would have to fly slower resulting in longer lap times. Similarly, the Clark Y airfoil would also generate more lift than necessary at the target cruise velocity. While the NACA 2412 did not have the highest lift coefficients, it was found to provide sufficient lift at T-O and appropriate lift to meet cruise requirements. The NACA 2412 also had a drag bucket which comparable to the Clark Y. Thus, the NACA 2412 airfoil was chosen because it had the lowest C_{Dmin} , a wide drag bucket, and the desired lift characteristics at the target cruise velocity.

4.4.2 Empennage Sizing

Stabilizer airfoils are generally symmetric as they may either provide lift or downforce, depending on the final trim state of the aircraft. WUDBF selected the NACA 0010 airfoil as it is common for stabilizers and used successfully in past WUDBF projects. Initial sizing of the empennage for stable flight began by investigating the vertical tail and horizontal tail volumes for aircraft of the same class. For a single-engine aircraft, the initial estimate was set at around $V_{HT} = 0.80$ and $V_{VT} = 0.04$ [2]. Initial horizontal and vertical tail areas were then solved for using Eq. 4.6 and Eq. 4.3 [2]:

$$S_{HT} = \frac{V_{HT} \cdot S_{ref} \cdot c_{ref}}{I_T} \quad (4.2)$$

$$S_{VT} = \frac{V_{VT} \cdot S_{ref} \cdot b_{ref}}{I_T} \quad (4.3)$$

Where S_{HT} is the horizontal tail area (cm²), V_{HT} is the horizontal tail volume, S_{VT} is the vertical tail area (cm²), V_{VT} is the vertical tail volume, S_{ref} is the main wing area [cm²], c_{ref} is the geometric chord (cm), b_{ref} is the span of the main wing [cm], and I_T is the optimal tail arm length (cm). Based on these calculations, the VT and HT planform areas were set at 387.5 cm² and 1296 cm² respectively. An initial assumption was made that the AR of the HT would be equal to about half the AR of the main wing and that the AR of the VT would be about 2.5 [2]. Initial estimates for the span of the VT and HT were then made using Eq. 4.4 and Eq. 4.5 as shown below [2]:

$$b_{HT} = \sqrt{AR_{HT} \cdot S_{HT}} \quad (4.4)$$

$$b_{VT} = \sqrt{AR_{VT} \cdot S_{VT}} \quad (4.5)$$

Where b_{HT} is the horizontal tail span (cm), and b_{VT} is the vertical tail span (cm). Figure 4.4 shows the preliminary configuration of *Ursa Major* compared to the idea aircraft used by Gudmundsson in his calculations.

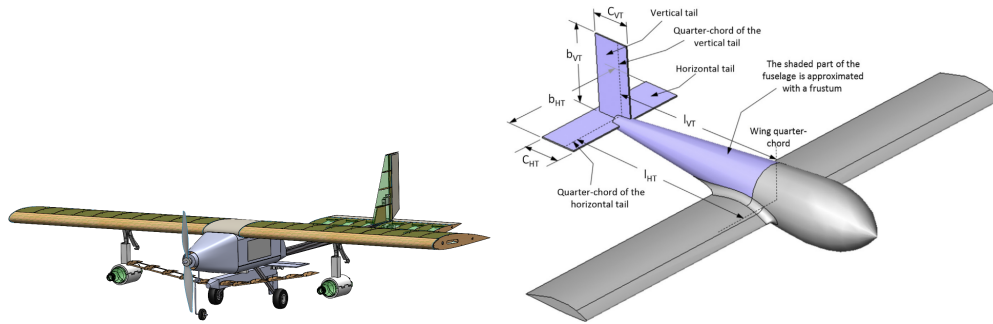


Figure 4.4: Initial configuration of the Ursa Major(left) vs. simplified aircraft layout used by Gudmundsson (right) [2].

4.4.3 High Lift Devices

It was decided that there was no need for high lift devices on the ends of the wing. The cost in weight and drag that high lift devices would outweigh the benefit in additional lift that they would give the aircraft.

4.4.4 Control Surface Sizing

Given the traditional design of this aircraft, all control surfaces (rudder, elevator, and ailerons) were designed using rule-of-thumb parameters from Gudmundsson [2]. Control surface sizing was verified using Eq. 4.6 from Gudmundsson [2], previous WUDBF aircraft, and prototype performance.

$$C_{l_{\delta\alpha}} = \frac{c_{l_{\delta\alpha}} c_R}{Sb} \left[(b_2^2 - b_1^2) + \frac{4(\lambda - 1)}{3b} (b_2^3 - b_1^3) \right] \quad (4.6)$$

Where $C_{l_{\delta\alpha}}$ is the aileron authority, $c_{l_{\delta\alpha}}$ is the change in lift coefficient with aileron deflection, b is the wing span [m], S is the wing area [m²], c_R is the root chord [m], and λ is the taper ratio. Plain flaps were chosen to simplify the manufacturing process. The hinge point of the flaps was placed at the center of the leading edge arc to minimize large gaps and overhangs when the surfaces are deployed. The flaps were given a chord fraction of 30%, and a span of 60% of the wing. The ailerons had a chord fraction of 30% and a span of 40% of the wing.

Both the flaps and the ailerons were designed to deploy between $\pm 30^\circ$. Using data taken from XFLR5, Fig. 4.5 shows the plotted lift coefficient versus AoA with the flaps deployed at 30° and with no flaps. Chord fractions for the control surfaces on the empennage were 30% for the elevator, and 40% for the rudder. The rudder was designed to deflect a maximum of 30° , while the elevator was designed to deflect a maximum of 30° .

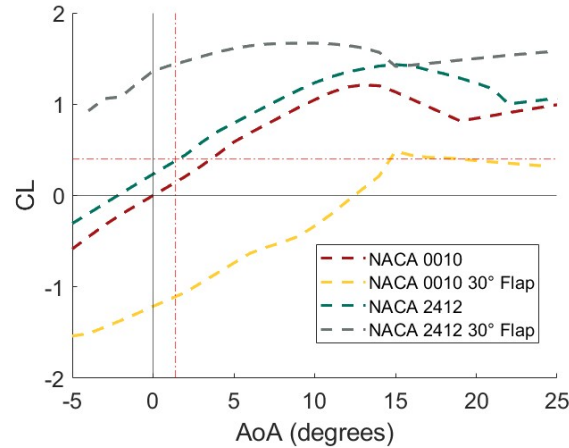


Figure 4.5: C_L of regular and flapped airfoils.

4.4.5 Static Stability Analysis

The static stability of the aircraft is important because it will prevent the aircraft from stalling and crashing, as well as reorienting the aircraft to fly at the optimized pitch angle of 0° . Good static stability will also help stabilize the aircraft against sudden wind gusts parallel to the flight path.

To verify the static stability of the aircraft design, the static margin was calculated using Eq. 4.7.

$$SM = \frac{X_{NP} - X_{CG}}{MAC_{Wing}} \times 100\% \quad (4.7)$$

Where SM is the static margin [%], X_{NP} is the neutral point of the aircraft in the x (longitudinal) direction [cm], X_{CG} is the center of gravity of the aircraft in the x direction [cm], and MAC_{Wing} is the mean aerodynamic chord of the wings [cm]. X_{NP} , X_{CG} , and MAC_{Wing} were automatically calculated by XFLR5. The static margin at cruise for M1 is 15%, and 7% for M2 and M3. This passes for static stability as a static margin between 5% and 15% is recommended [8]. The nature of the static margin and how it changes at extreme angles is shown in Fig. 4.6. It shows that the static margin (proportional to the slope of the lines) is relatively constant at all angles, except for M2 and M3 past -10° . This is acceptable as -10° is an extreme angle that is unlikely to be used, as the expected landing angle is approximately -7° .

4.4.6 Dynamic Stability Analysis

The dynamic stability characteristics of the aircraft were analyzed using XFLR5. The Root Locus plot below in Fig. 4.7 shows the dynamic stability modes for all missions. Mission 2 stability is identical to Mission 3 stability before dropping the glider. Mission 3 stability is shown after dropping the glider.

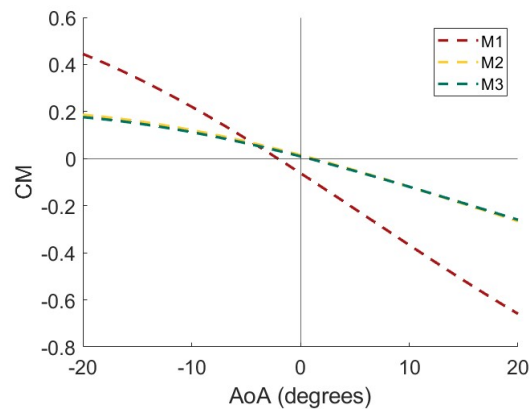


Figure 4.6: Static stability representation using plot of C_M vs. AoA.

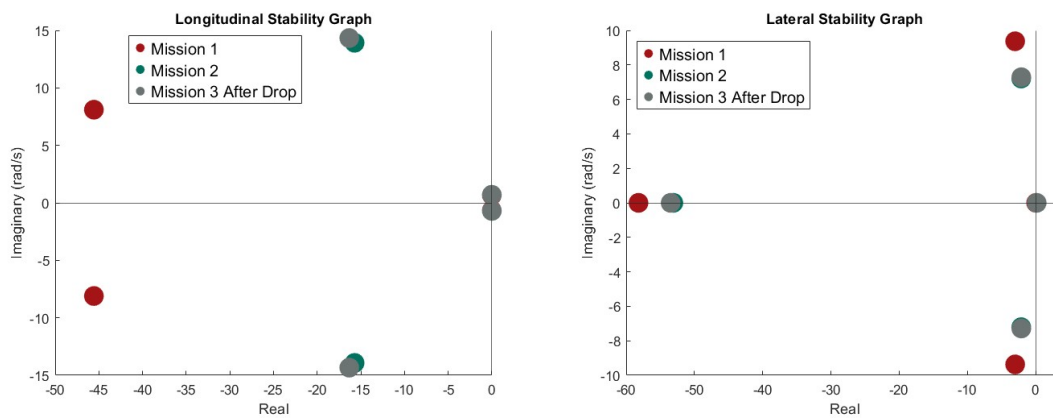


Figure 4.7: Root locus longitudinal plot (left) and lateral plot (right).

Based on the root locus plots, all modes were stable for both loading conditions except for Mission 1 Dutch Roll damping; this is acceptable because Mission 1 does not require a high degree of stability and it has acceptable values for stabilizing frequencies. Stability was prioritized for Mission 2 and Mission 3 because more score could be achieved during those missions. The design goal of this aircraft was to satisfy the level 1 flying qualities of a Class 1 aircraft described in MIL-F-8785C [9]. The dynamic stability characteristics from XLFR5 for all missions are shown in Tables 4.3, 4.4, and 4.5.

Table 4.3: Dynamic stability table M1.

Mode	ζ	$\omega_n(\text{rad/s})$	$\zeta\omega_n(\text{rad/s})$	τ (s)
Short Period	0.978 ($0.35 < \zeta < 2.00$)	46.28	8.11	-
Phugoid	0.025 ($\zeta > 0.004$)	0.584	0.584	-
Roll	-	-	-	0.017 ($\tau < 1.4$)
Dutch Roll	0.025 ($\zeta > 0.17$)	9.85 (> 1.0)	9.39 (> 0.35)	-
Spiral	-	-	-	9.379

Table 4.4: Dynamic stability table M2.

Mode	ζ	$\omega_n(rad/s)$	$\zeta\omega_n(rad/s)$	$\tau (s)$
Short Period	0.749 ($0.35 < \zeta < 2.00$)	21.00	13.93	-
Phugoid	0.019 ($\zeta > 0.004$)	0.672	0.672	-
Roll	-	-	-	0.019 ($\tau < 1.4$)
Dutch Roll	0.276 ($\zeta > 0.17$)	7.54 (> 1.0)	7.25 (> 0.35)	-
Spiral	-	-	-	4.447

Table 4.5: Dynamic stability table M3 after drop.

Mode	ζ	$\omega_n(rad/s)$	$\zeta\omega_n(rad/s)$	$\tau (s)$
Short Period	0.752 ($0.35 < \zeta < 2.00$)	21.73	14.34	-
Phugoid	0.019 ($\zeta > 0.004$)	0.666	0.666	-
Roll	-	-	-	0.019 ($\tau < 1.4$)
Dutch Roll	0.272 ($\zeta > 0.17$)	7.61 (> 1.0)	7.32 (> 0.35)	-
Spiral	-	-	-	4.649

With the exception of the phugoid damping ratio and the short period natural frequency, all modes satisfied the MIL-F-8785C level 1 requirements [9]. The phugoid damping ratio did satisfy MIL-F-8785C level 2 requirements and pilot evaluation during. The short period frequency fell just outside of level 2, but satisfied pilot evaluation.

4.4.7 Drag Analysis

Ansys returned a force value for drag. The drag coefficient was then calculated using Eq. 4.8:

$$C_{dmin} = \frac{2 \cdot F}{\rho \cdot u^2 \cdot A} \quad (4.8)$$

where F is the simulation drag force on that surface, ρ is the air density, u is the cruise velocity, and A is the area of the wings. This yielded a total parasitic drag coefficient of 0.0817. The complete breakdown of parasitic drag is presented in Fig. 4.8. The component drag build-up method (CDBM) as described by Gudmundsson [2] was not used because it had given inaccurate numbers in previous WUDBF designs. The CDBM method predicted a significantly lower drag coefficient than the flight testing results for this year and previous years. The combined drag force from parasitic effects and lift during cruise is expected to be 45.2 Newtons. The parasitic drag force is expected to be 40.0 Newtons.

The wings were the largest contributor to parasitic drag for all mission configurations, which is good, because it means the stabilizers and other components that do not contribute to cargo capacity are not overly significant sources of drag. The horizontal stabilizer took up a much larger amount of drag than was expected, but it serves to increase the stability of the plane.

The parasitic drag was then added to the lift-induced drag, calculated using XFLR5, to give the total drag polar, as shown in Fig. 4.9.

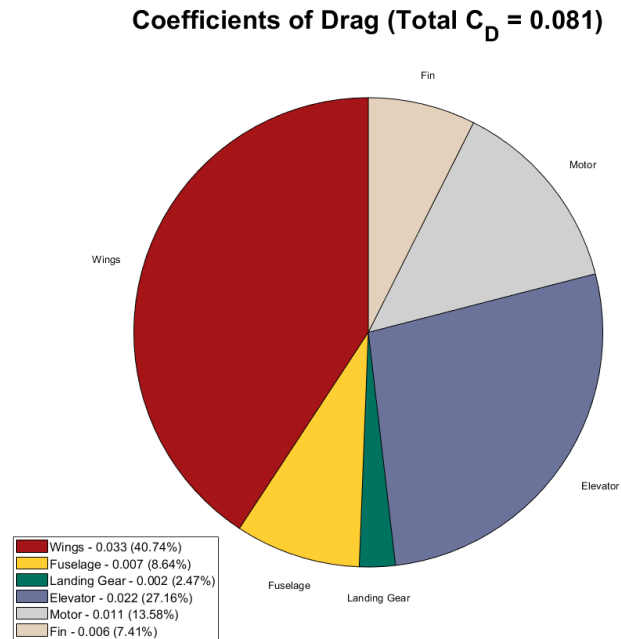


Figure 4.8: Breakdown of parasitic drag by component.

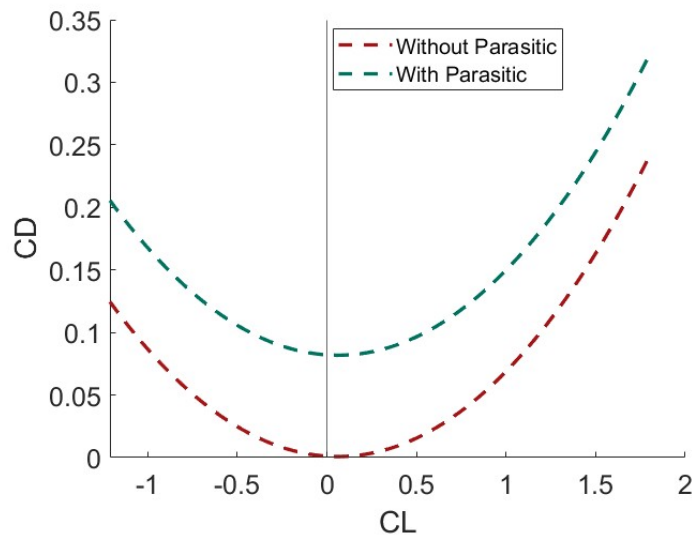


Figure 4.9: Aircraft C_D vs. C_L .

The glider and pylons are expected to add 0.006 to the total drag coefficient based on preliminary simulations.

4.4.8 Glider Performance

The performance of the glider depends on its dynamic stability and its glide ratio which can be determined from the CL/CD . These parameters were all determined in XFLR5.

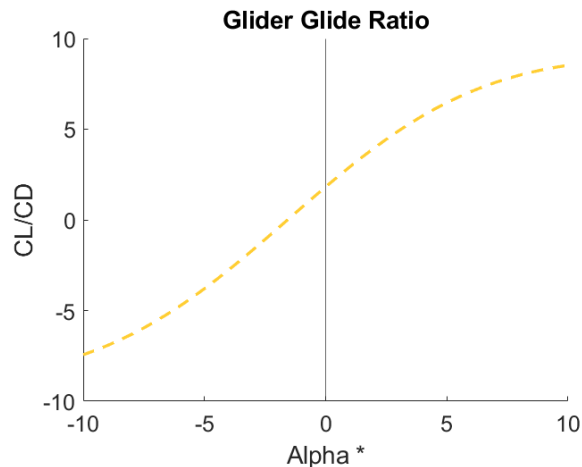


Figure 4.10: Glider glide ratio.

Given the estimated drop height, WUDBF has selected an angle of attack of 5 degrees to give us a lift to drag ratio of approximately 6 in order to provide enough range to reach the designated landing zone as shown in Fig 4.10.

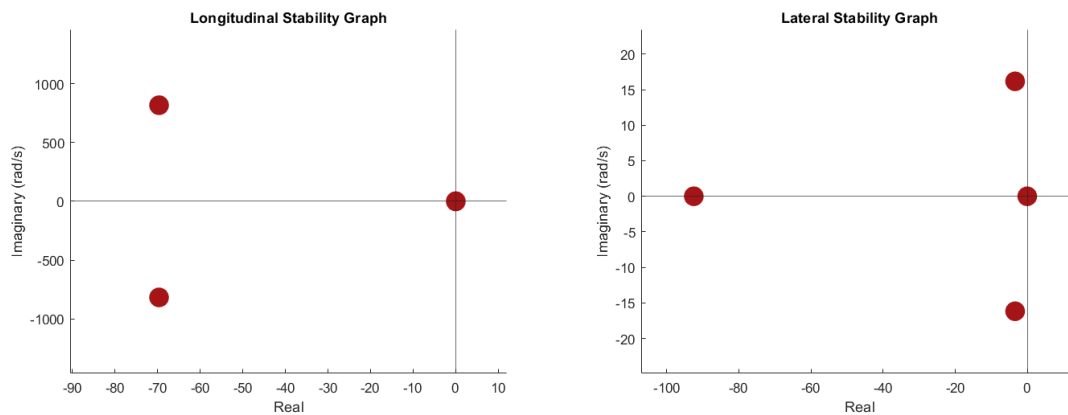


Figure 4.11: Root locus longitudinal plot (left) and lateral plot (right).

The glider is well above the parameters for lateral and longitudinal stability for a level 1 Class 1 aircraft, as described in MIL-F-8785C [9]. The dynamic stability can be seen in Fig. 4.11. Dynamic stability is very important because the glider must be stable without any pilot input.

The glider's autonomous function will be controlled by QGroundControl software being uploaded to an onboard pixhawk. The control surfaces will be a rudder and an elevator. The glider is planned to go into a spinning landing once the glider is detected to be over the center of the landing area. The programming and controls are modeled in Fig. 4.12 in Simulink [10].

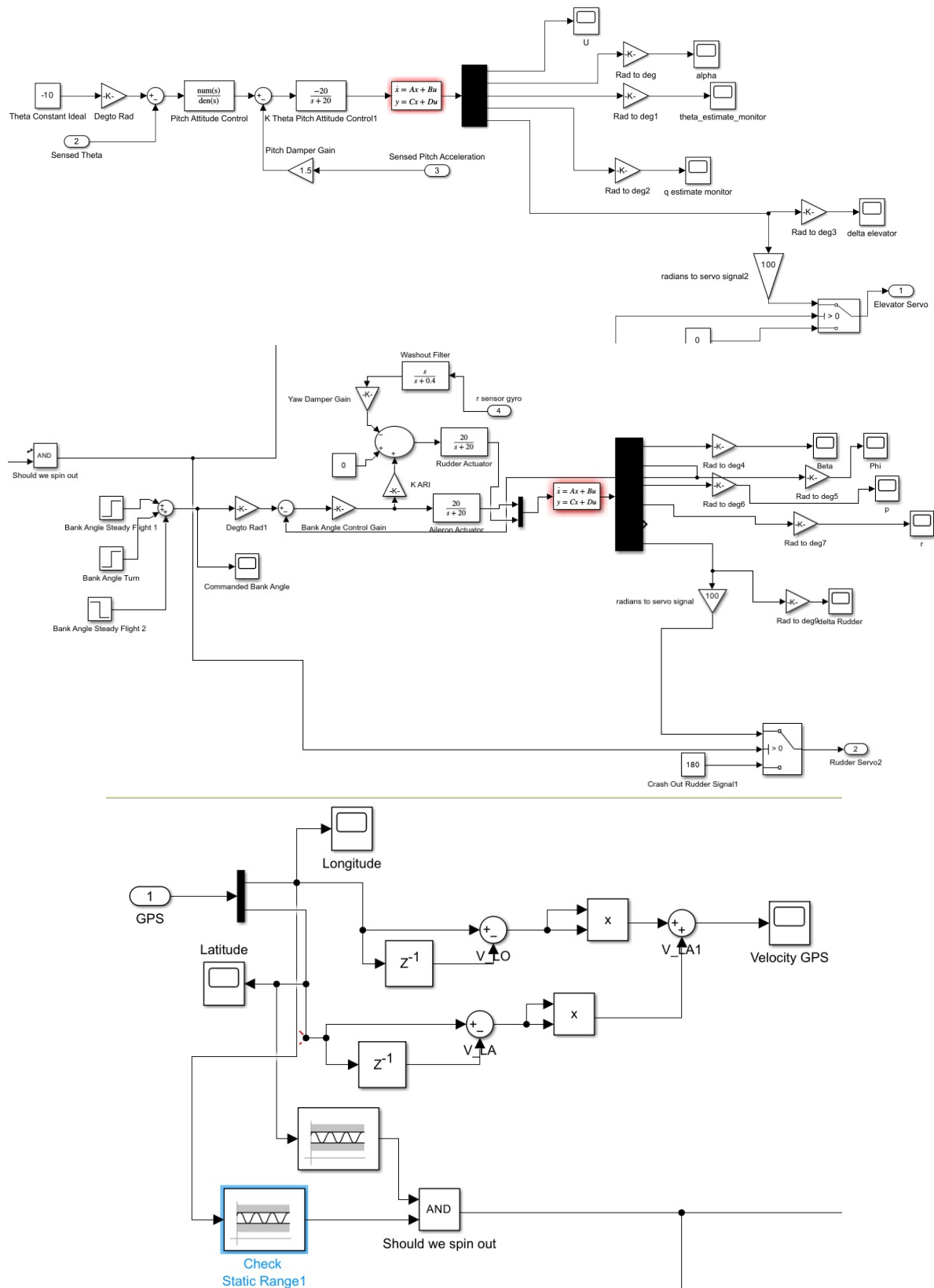


Figure 4.12: Longitudinal control (top), lateral control (middle), GPS signal (bottom).

4.5 Propulsion

The propulsion system was designed taking into account the results from the performance optimizer as stated in Table 3.3. In particular, the specific requirements that were prioritized are as follows; a required T/W of 0.950 for T-O due to the high load payload expected of M2, a required electrical power of 1560 W and 1979 W for T-O and cruise, respectively (see Section 4.2), a desired cruise velocity of 31.3 m/s (see Table 3.3), and enough endurance to complete at least three laps for M1 and M2. Additional considerations included maximizing efficiency, especially during cruise, and minimizing overall propulsion system weight due to already high payloads.

4.5.1 Battery Selection

The DBF rules allow for the use of either LiPo (Lithium Polymer) or NiMH (Nickel-Metal-Hydride) batteries for propulsion. While NiMH batteries tend to be more robust and safe, LiPo batteries offer higher energy density, power to weight ratio, discharge rate, efficiency, and reliability. LiPos also tend to take up less space for a given energy capacity. Thus, WUDBF chose to use LiPo batteries for the propulsion pack.

LiPo's are commonly available in 3s, 4s, 5s, 6s, and 8s packs. For an aircraft of this size, LiPos below 6s are unlikely to provide the performance required. For 10s LiPos, there are not many (if any at all) available below the 100 Wh (2777.78 mAh for 10s) limit. However, for larger and heavier aircraft you need a larger motor/propeller, which requires a higher voltage to operate meaning a larger cell count. Therefore, the propulsion team chose 8s LiPo batteries over 6s for two main reasons: higher power available at takeoff and lower amperage draw. First, the 8s battery has higher voltage than 6s, 29.6V vs. 22.2V respectively, which allows the system to draw less current for the same power output. This is due to the direct relationship between power (P), voltage (V), and current (I), as given by Eq. 4.9.

$$P = V * I \quad (4.9)$$

Lowering the amperage draw is favorable as it lead to less heat being generated by the system. This is explained by Joule's first law, which states that the heat generated in a system is proportional to the square of the current, the resistance of the wire, and the time the current flows. The decrease in heat generation leads to increased efficiency as less energy is turning to heat and more is going to power the system. Second, the higher voltage of an 8s battery allows for higher available thrust and speed when compared to a 6s battery of similar watt-hours. A motor's no-load revolutions per minute (RPM) is given by Eq. 4.10.

$$RPM = Kv * V \quad (4.10)$$

where Kv is the motor's rotations per voltage supplied (generally given by the manufacturer). So on the same motor and same propeller, an 8s battery could produce higher RPM's than a 6s battery, and thus making more thrust and speed available.

For the battery selection, WUDBF chose to go with the largest capacity battery allowed. The MaxAmps 3250 mAh 8s battery pack was chosen to run the propulsion system because it met the capacity requirement and was readily available. Thus, WUDBF chose to use 8s LiPos, since they are readily available below the 100 Wh (3378.38 mAh for 8s) limit and have been used in the past to power competition aircraft of a similar size to this year's design. Additionally, the battery also possessed high discharge rates of 23C-Constant and 150C-Burst, which allows for a stronger throttle punch and improved takeoff performance.

4.5.2 Motor and Propeller Selection

In order to identify possible motor-propeller combinations, WUDBF employed the online propulsion design tool eCalc [11]. Using the parameters listed in the table below, WUDBF searched through the eCalc database for motor-propeller combinations that met the stated performance requirements.

Table 4.6: eCalc search parameter.

Parameter	Value	Reasoning
T/W	0.94	See Section 4.2
Max Cruise Velocity [m/s]	31.3	See Section 3.3.1
Endurance [min]	5	Flight time window
Battery	8s, 3250mAh	See Section 4.5.1
Max Prop Diameter [in]	20	Estimated max ground clearance

From this search, a list of over 50 unique motor-propeller combinations were identified. This list was then further refined by limiting the possible motors to only those from trusted manufacturers (based on past experience and general online consensus) that were available, in-stock, at the time of the search. This narrowed the possibilities down to the four motor-propeller combinations shown in Table 4.7:

Table 4.7: Possible motor-propeller combinations.

Outputs	Scorpion A-4225-500 V2	Scorpion A-4225-500	Sunny sky X4120-480	Sunny sky X4125-480
Propeller	15x10	14x12	15x10	15x10
Max Velocity (m/s)	36.11	38.61	34.44	35.28
Mixed Flight Time (min)	2.6	2.8	2.8	2.7
RPM (static)	10373	10571	9935	10188
T/W	1.17	1.18	1.07	1.13
Static Thrust (kgf)	9.354	9.493	8.581	9.024
Current @ Max (A)	111.8	106.1	99.2	104.6
Drive Weight (kg)	1.273	1.273	1.156	1.225

All four choices met or nearly met all of the stated requirements. Thus, in order to further evaluate each combination, plots of electrical efficiency vs. throttle, endurance vs. speed, endurance vs. throttle, and thrust vs. electrical power were generated using data from eCalc:

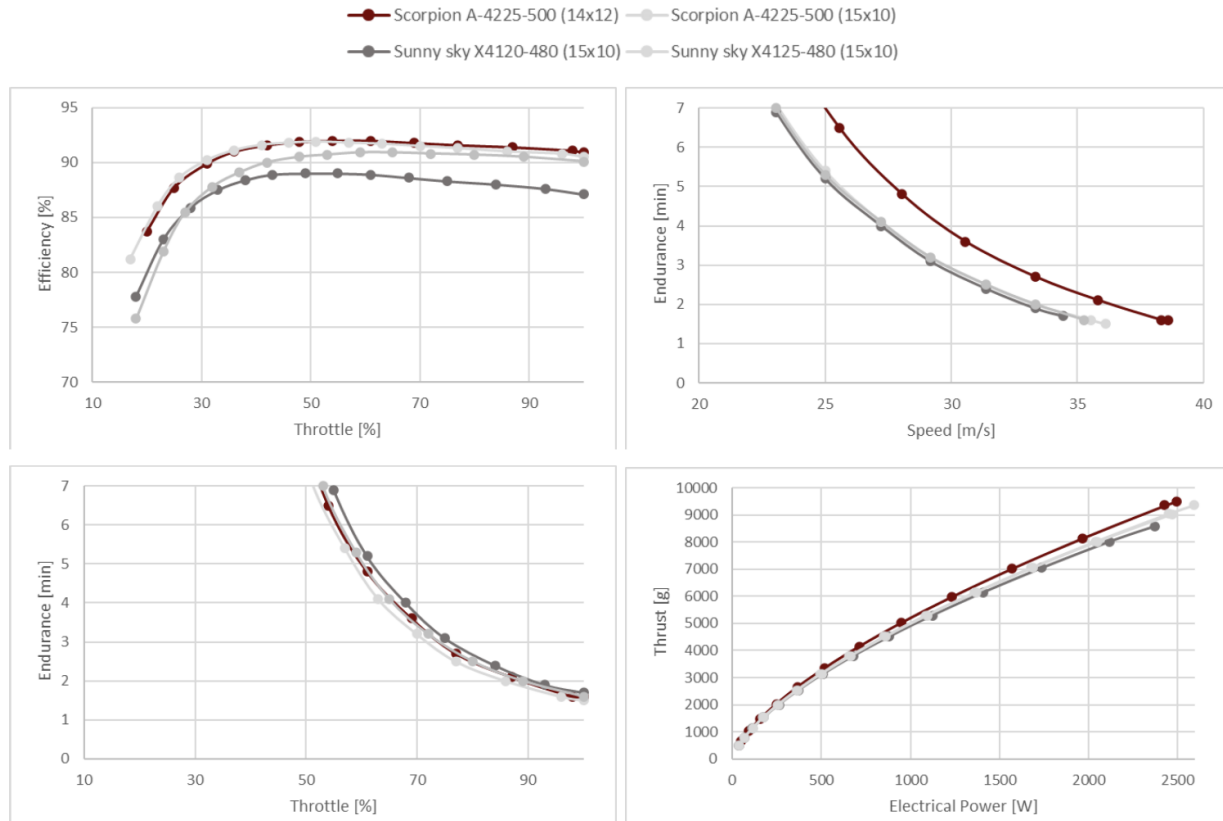


Figure 4.13: Predicted performance data for motor-propeller combinations.

From Figure 4.13, the Scorpion A-4225-500 motor with a 14x12 propeller had the greatest electrical efficiency over relevant throttle settings (60-100%). This motor-propeller combination also provided superior thrust-to-power performance and endurance that was comparable to the other choices. From Table 4.7, the Scorpion A-4225-500 motor did have the highest drive weight, but it still provided the greatest endurance. Based on these factors, the Scorpion A-4225-500 motor with a 14x12 propeller was ultimately chosen for the aircraft's propulsion system.

4.5.3 ESC and Fuse Selection

Given the predicted max current of 111.8 for the chosen propulsion system, WUDBF chose to use the Phoenix Edge Lite 130A (8S) as the ESC. The Phoenix Edge Lite can handle 130A continuous current, which provides an adequate safe margin. Additionally, the electronic speed controller (ESC) can handle the increased voltage from the 8S LiPo battery present in the propulsion system.

4.5.4 Refinement for M2

Given that the propulsion system was designed primarily with M3 in mind, WUDBF next considered whether it could be refined to improve M2 performance, since the competition rules allow for the propeller and battery to be swapped out for different missions.

Since M2 has a greater emphasis on speed rather than efficiency, WUDBF considered using a larger propeller with a higher pitch. The larger diameter allows the propeller to move more air, thereby generating greater thrust, allowing for an increase in max velocity. Additionally, the higher pitch moves more air backwards per revolution, contributing to top speeds. The main downside of using larger propellers with a higher pitch is the higher power draw from the motor, leading to lower flight times. Also, higher pitch propellers sacrifice efficiency at lower

airspeeds, negatively impacting static thrust and takeoff performance [12]. Because of concerns for the motor power limit, propeller RPM limits, and takeoff distance, WUDBF could not find a propeller that increased cruise velocity without decreasing the static T/W and/or endurance below acceptable levels. Thus, the 14x12 propeller was selected for use in all missions.

4.5.5 Required vs. Available Thrust Analysis

The dynamic performance of the propulsion system was evaluated to verify eCalc's predicted performance and to determine the aircraft's speed envelope. eCalc does not output dynamic thrust data so this data was instead used from the APC Propellers website (the chosen propeller manufacturer) [13]. The APC performance data was generated based on vortex theory using actual propeller geometry and the NASA Transonic Airfoil Analysis Computer Program [14] to estimate section lift and drag. Predicted RPM values from eCalc were used to query the APC dataset, with the corresponding dynamic thrust values calculated using an interpolation script written in MATLAB.

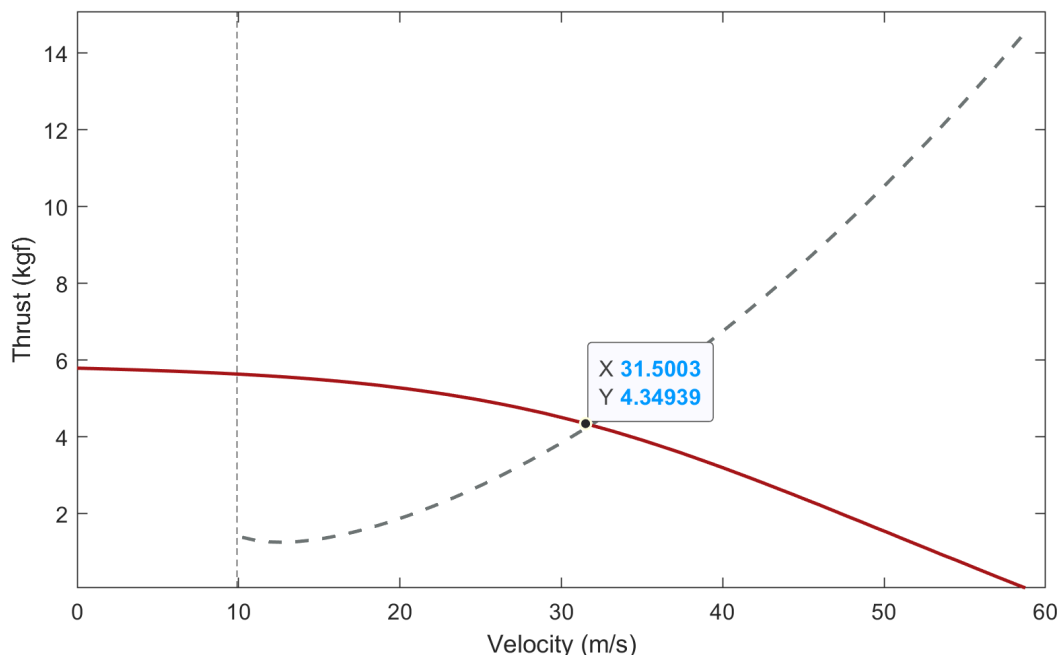


Figure 4.14: Required vs. available thrust curves.

Based on the analysis, the estimated maximum velocity for the aircraft is 31.5 m/s. The velocity calculated above may differ from the value given by eCalc because propCalc (the calculator used to acquire the values from Table 4.7) uses multiple simplifications and assumptions for the drag on the aircraft, which may lead to a difference in the calculated speed vs. the actual speed. The analysis done above more accurately takes into account the drag on the aircraft and thus has a more accurate estimate for the cruise speed of the aircraft.

4.6 Predicted Aircraft Performance

In order to integrate the various aspects of the preliminary design and predict overall aircraft performance, WUDBF utilized Km_sim, an in-house kinematic mission simulator written in MATLAB [15].

4.6.1 Brief Explanation of Km_sim

Km_sim works by simulating an aircraft's performance as it flies through a mission profile. Each mission profile consists of a series of functions based on simple kinematics and Newton's second law describing the motion of

the aircraft during a particular phase of flight (e.g., takeoff, climb, cruise, etc.). Specific mission profiles were written for M1, M2, and M3. Aerodynamic and propulsion data from Section 4.4 and Section 4.5, respectively, were used to generate aerodynamic and propulsion performance look-up tables. These tables allow the simulation to determine the lift, drag, and thrust of the aircraft at any particular time point. During a simulation, Km_sim steps through each mission phase, numerically integrating the relevant equations of motion using ode45 and the look-up tables, while keeping track of changes in the state variables. For example, one mission phase is dedicated to calculating mission parameters during a constant radius turn. Fig. 4.15 contains sample code from this mission phase, in which lift is calculated using equations for centripetal acceleration, force, and lift.

```
% determine lift for a constant radius turn
accel_c = v^2/turn_radius;      % centripetal accel. in m/s^2
force_c = mass*accel_c;         % F=ma
lift     = sqrt(weight^2 + force_c^2)-thrust*sin(aoa);
```

Figure 4.15: Sample code from Km_sim, in which lift is calculated during a constant radius turn.

At the end of the simulation, plots are generated displaying how the state variables change over time as shown in Fig. 4.16 and Fig. 4.17. For a complete explanation of Km_sim and the source code see [15].

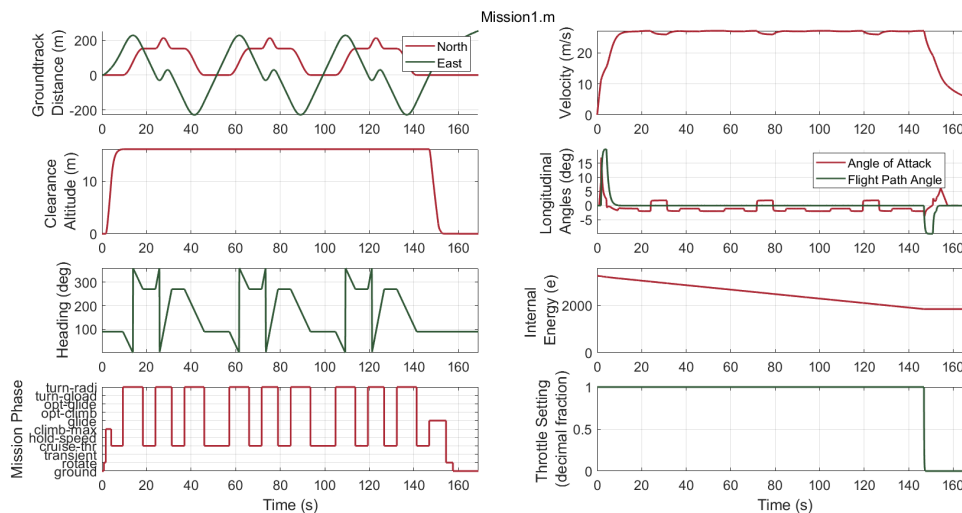


Figure 4.16: Plots of different state variable outputs versus time during a simulation.

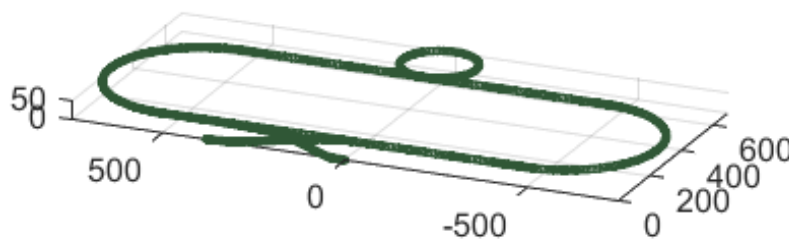


Figure 4.17: Trajectory plot (TOP) illustrating the path taken by the aircraft in 3D coordinates during a simulation.

4.6.2 Km_sim Results

Separate simulations for M1, M2, and M3 were carried out in Km_sim to estimate performance in competition. The results of this analysis are shown in Table 4.8. Normalized scores were calculated based on the maximum

competition scores predicted as part of the analysis in Section 3.3.1. For M1 and M3, a reduced cruise throttle was used, because M1 is a simple completion test and M3 is an endurance test. Simulations at full throttle for M3 showed that the battery would be depleted after around 3 minutes, long before the end of the 5-minute time window.

Table 4.8: Predicted aircraft performance.

	Mission 1	Mission 2	Mission 3
Inputs			
All-Up Weight	4.082 kg	7.134 kg	4.734 kg
X-1 Test Vehicle Weight	-	0.20 kg	0.20 kg
Fuel Tank Weight	-	2.50 kg	0.10 kg
Wing Loading	5.02 kg/m ²	8.78 kg/m ²	5.83 kg/m ²
$C_{L,max}$	2.42	2.42	2.42
Parasitic Drag	0.081	0.081	0.081
Cruise Throttle	0.8	1	0.8
Outputs			
Takeoff Distance	6.70 m	10.3 m	9.84 m
$C_{L,cruise}$	0.350	0.350	0.350
$C_{D,cruise}$	0.094	0.094	0.094
Cruise Velocity	27.17 m/s	33.12 m/s	27.10 m/s
Number of Laps	3	3	5 (Before X-1 launch)
Mission Time at End of Final Lap	2.32 min	1.95 min	4.74 min (At end of lap after X-1 launch)
Scoring			
Normalized Score	1	2	2.08 (0 bonus), 2.16 (1 bonus), 2.28 (2.5 bonus)

4.7 Uncertainties

The design tools and analysis methods used throughout the preliminary design process all possessed a certain level of uncertainty. The lift and induced drag of the aircraft were computed in XFLR5 using the vortex lattice method (VLM). Using VLM, lifting surface thickness and air viscosity were neglected. Additionally, the fuselage was excluded from the analysis. Based on previous WUDBF designs, however, it was determined that the decreased lift due to neglecting viscosity is approximately equal to the lift gained by excluding the fuselage. It has also been verified that the contribution of drag associated with wing thickness was small compared to total drag calculations. Therefore, the lift and induced drag of the actual aircraft was expected to be approximately equal to the XFLR5 predicted lift using the VLM. The parasitic drag estimate using Gudmundsson's method was also subject to a certain level of uncertainty, but that uncertainty was mostly accounted for in the CFD analysis. eCalc and the APC propeller performance database were the two primary tools used for propulsion design. These third-party tools both carry a certain level of uncertainty owing to the fact that it is not entirely clear how they calculate their outputs. However, this uncertainty is mitigated by the fact that this method of propulsion design has been used successfully by both WUDBF and other DBF teams in the past with an acceptable level of accuracy. Finally, Km.sim, while a very useful tool, makes a number of assumptions in its performance calculations. For instance, the simulation assumes that the pilot flies the aircraft on a perfect trajectory. The effects of wind are

also neglected in the simulation. Since Km_sim relied on data from the aerodynamic and propulsion designs, any uncertainties from those processes were carried over into the simulation. Ultimately, the nature of the iterative process meant that the uncertainty associated with any particular design tool was mitigated by ensuring that every aspect of the design agreed with the rest.

5 Detail Design

5.1 Dimensional Parameters

Following the preliminary stage, the design and geometry of the aircraft were finalized during the detail design stage. Table 5.1 summarizes the key dimensions of the final aircraft design.

Table 5.1: Key dimensions of final aircraft design.

Vertical Stabilizer		Horizontal Stabilizer	
Span (mm)	250	Span (mm)	540
Mean Chord (mm)	155	Mean Chord (mm)	240
Area (m ²)	.04	Area (m ²)	0.13
Aspect Ratio (AR)	0.48	AR (Aspect Ratio)	2.25
Wing		Overall Aircraft	
Span (mm)	1830	Length (mm)	1668.78
Mean Chord (mm)	444	Width (mm)	1828.8
Area (m ²)	0.74	Empty Weight (kg)	4.501
Aspect Ratio (AR)	3.776	Max Weight (kg)	8.147

5.2 Structural Characteristics and Design Methodology

WUDBF's primary structural design goals were to maximize strength and stability, improve accessibility to electronics and payload compartments, minimize weight, and simplify the manufacturing process. The overall structure is designed to direct loads from each subsystem into the major load-bearing components of the aircraft. The fuselage consists of a fiberglass monocoque body with a carbon fiber boom running from nose to tail. Propulsive loads from motor thrust and torque that can cause bending, torsion, and vibrations are directed through a wooden motor mount and distributed into the fuselage. Ground loads from the landing gear are directed through the fuselage structure and into the central CF rod. These load paths are illustrated in Fig. 5.1. The wing consists of wooden ribs connected by a stringer and 2 CF spars. One runs through the entire wing, and the other spans through the center 14 ribs. Both provide bending and torsional stiffness. Aerodynamic loads from the wing are transferred to the wing box, where they are then distributed across the fuselage structure and the central CF rod. The empennage is directly connected to the central CF rod via a CF spar and locking wood structure, distributing loads into the central rod and the fuselage.

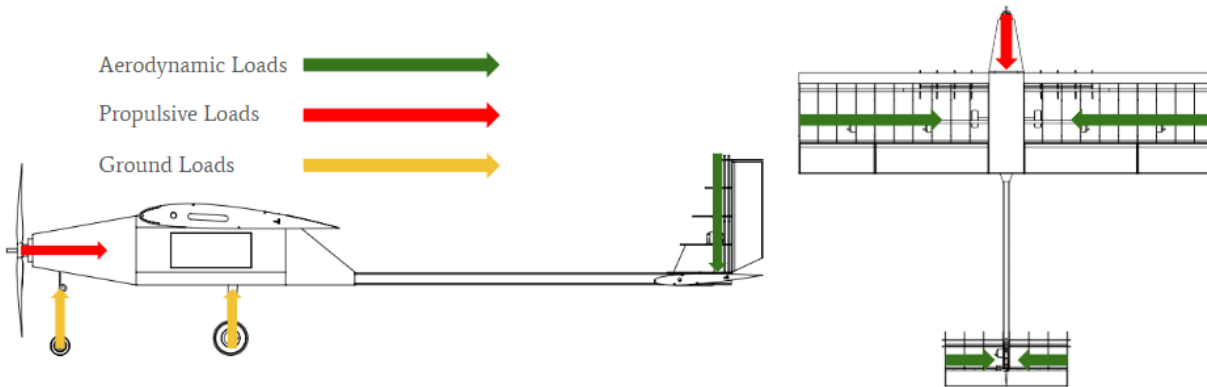


Figure 5.1: Load path diagram.

5.3 Subsystem Design

5.3.1 Fuselage Design

The fuselage is a fiberglass monocoque shell, with birch plywood firewalls glued inside. It is primarily designed to encase the electronics package, control electronics, and battery. It also directly connects the motor, landing gear, wing, and tail. This design can be seen in Fig. 5.2. Birch plates connect to the motor mounting plate and form shelves for the electronics and battery. An 7-inch-wide opening is cut into the side of the fuselage. A hinged hatch covers this opening and is secured with plastic latches. This facilitates easy access to electronics for maintenance or component swaps. A CF tail boom spans the entire length of the fiberglass fuselage and extends to the empennage. The top of the fuselage is shaped to precisely align the wing at the proper angle. A birch wood plate at the top of the fuselage, supported by the firewalls, reinforces the wing mount and helps transfer its loads to the fuselage.

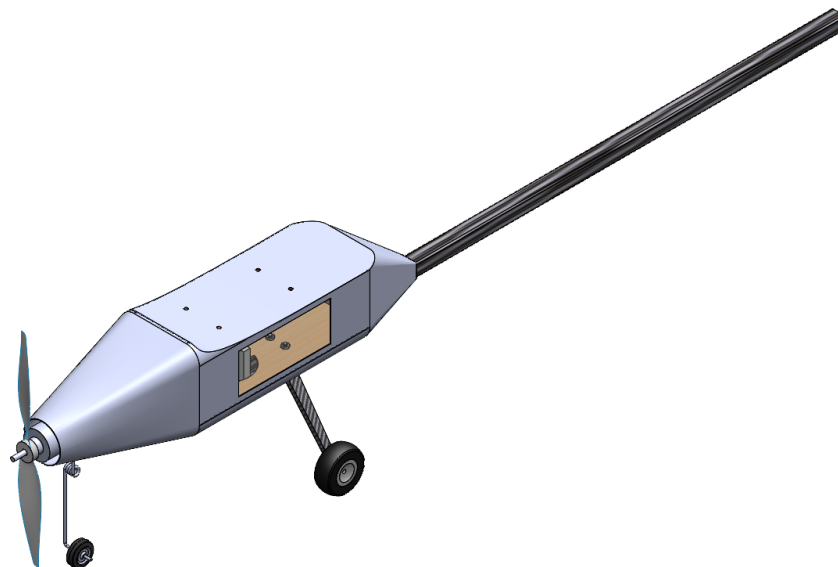


Figure 5.2: Fuselage.

5.3.2 Wing Design

The wing structure consists of 22 ribs and two CF spars. One CF spar runs through the entire wing, while the other supports the middle 14 ribs. This optimizes bending stiffness and reinforces the anchor point to the fuselage, also while reducing unnecessary weight. Both CF spars connect to a wing box, providing structural support and reducing wing torsion. Two bolts pass through each CF spar, fastening the wing to the birch plate in the fuselage. The wing structure includes servo and control surface attachment points. The control surfaces are constructed from XPS foam with a fiberglass layup. Balsa wood reinforcements wrap the leading edges of the ribs, enhancing wing stiffness. The wing assembly, including ribs, CF spars, leading edge, and control surfaces, is shown in Fig. 5.3.

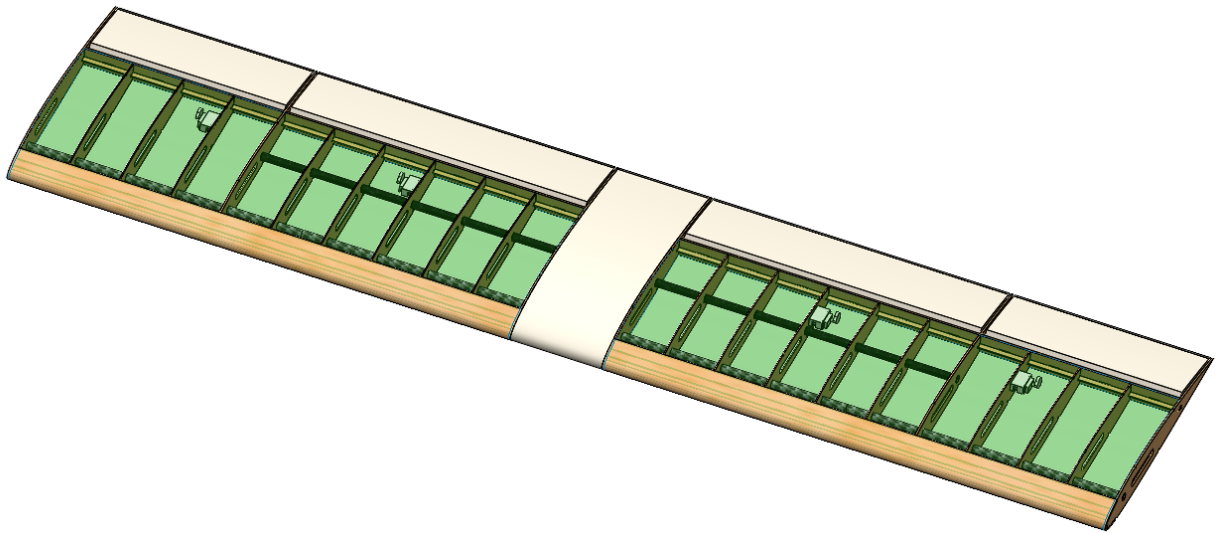


Figure 5.3: Wing assembly.

5.3.3 Empennage Design

The empennage was designed similarly to the main wing. The horizontal stabilizer is supported by a single CF rod through the entire span. The vertical stabilizer does not have a CF rod. Instead, a plywood tab extends through its lower ribs and fits around the horizontal CF rod. Both the horizontal and vertical stabilizers are glued to the carbon fiber tail boom coming from the fuselage, increasing its stiffness. Mounting plates for control servos, as well as hinges for the rudder and elevator, are also integrated into the rib structure. This structure can be seen in Fig. 5.4.

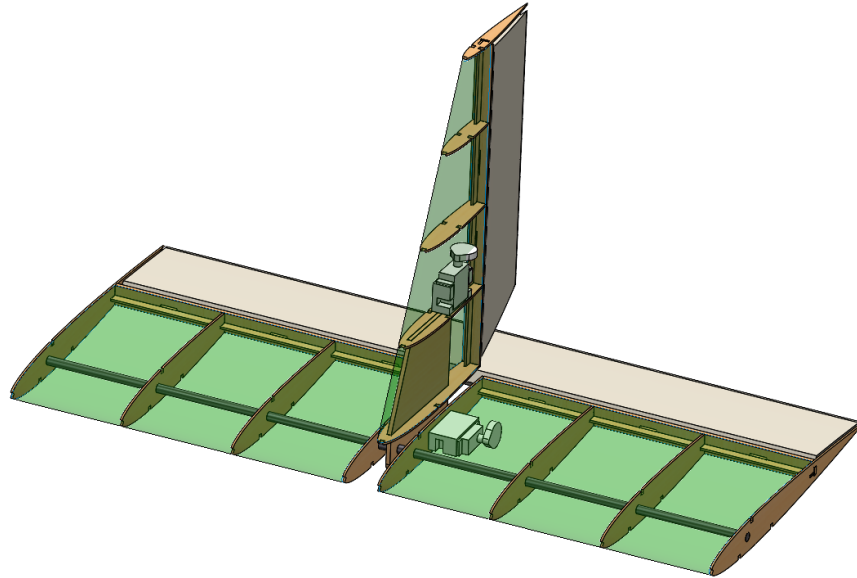


Figure 5.4: Tail assembly.

5.3.4 Landing Gear

CF landing gear was purchased online because WUDBF did not have the equipment to manufacture it. The JTC-S10 Bow Tricycle landing gear was selected and positioned in the rear of the fuselage. The landing gear is securely bolted to the CF boom at the underside of the fuselage, ensuring structural stability. The nose landing gear consists of a spring-wheel landing gear secured to the back of the motor mount. This structure is shown in Fig. 5.5.

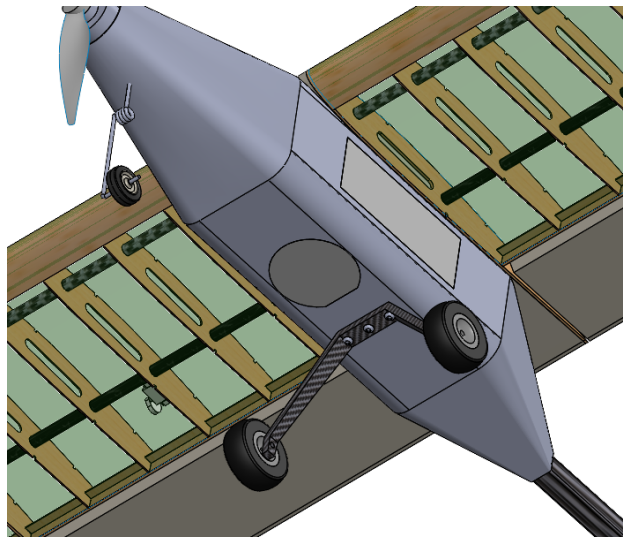


Figure 5.5: Landing gear configuration.

5.3.5 M2 Payloads

The payloads for M2 are the X-1 test vehicle and four externally mounted fuel tanks. The fuel tanks are 16 fl oz bottles with fuel, mounted by removable pylons, which are clamped securely to the CF spar from the underside of the wing through holes in the fiberglass. The X-1 test vehicle is mounted by the release mechanism, which locks the vehicle in place during the flight. This mechanism extends out the bottom side of the fuselage, through a hinged door secured with plastic latches.

5.3.6 M3 Payloads

The payloads for M3 are the X-1 test vehicle and two externally mounted fuel tanks. In this mission, the fuel tanks are empty. The glider is mounted by the release mechanism, which locks the vehicle in place until the pilot initiates the movement of the servo motor. The full assembly configuration for M2 and M3 is shown in Fig. 5.6.



Figure 5.6: M2 and M3 mission configuration.

5.3.7 Glider

The X-1 test vehicle was designed to minimize weight while maintaining sufficient maneuverability. The wings consist of 12 balsa wood ribs connected by balsa rods, providing structural integrity. The fuselage consists of balsa wood firewalls and stringers, with a fiberglass monocoque shell. A 2-inch-wide hole on the roof exposes a key slot made of balsa wood plates that interact with the release mechanism to secure the vehicle. The empennage was simplified further with flat horizontal and vertical stabilizers. In order to provide sufficient space between the main fuselage and the X-1 test vehicle the vertical stabilizer was built to extend downward. Control surfaces consist of flat XPS foam board control surfaces on the vertical and horizontal stabilizers that are controlled by servos glued to the fuselage. The X-1 test vehicle assembly is shown in Fig. 5.7.

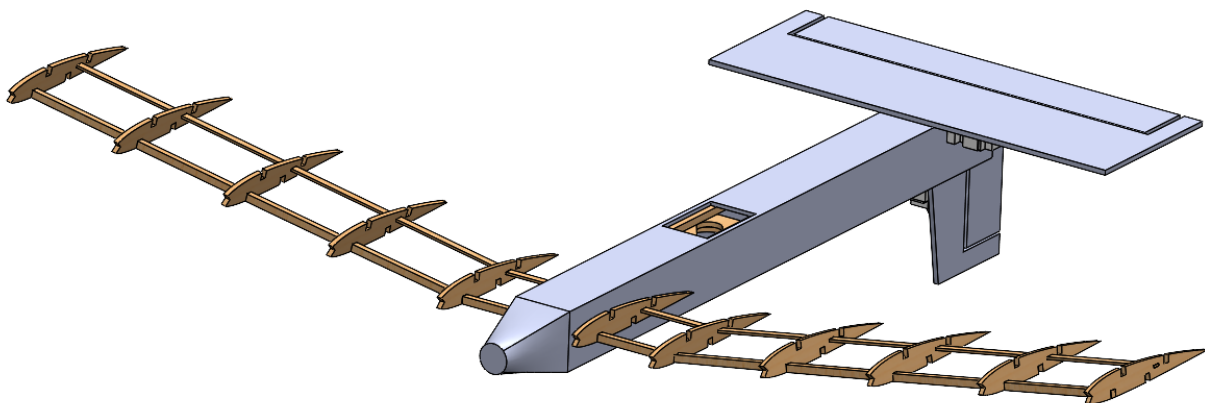


Figure 5.7: X-1 test vehicle assembly.

5.4 Pylons

The pylons were designed to ensure the simplicity of assembly consisting of two key components that work together to secure the fuel tanks. The first is a handheld spring clamp. The clamp is attached to the front CF spar through holes in the fiberglass of the underside of the wing between the 5th and 6th ribs of the wing. The second component is the 3D-printed payload retainer, which consists of two interlocking parts, a shaft and a hinged snap-fit retention shell. The shaft of the retainer has a contoured cavity designed to secure one handle of the handheld clamp. To ensure the retainer remains firmly attached to the clamp, the cavity is covered by a sliding panel that locks the assembly in place. The shaft also forms the upper section of the retention shell that anchors the fuel tank. It incorporates a hinge and two snap-fit receptacles to allow for simple assembly. The retention shell has a cylindrical geometry to match the fuel tank. Two snap-fit tabs join the assembly by locking into the corresponding receptacles on the shaft. This system is illustrated in Fig. 5.8.

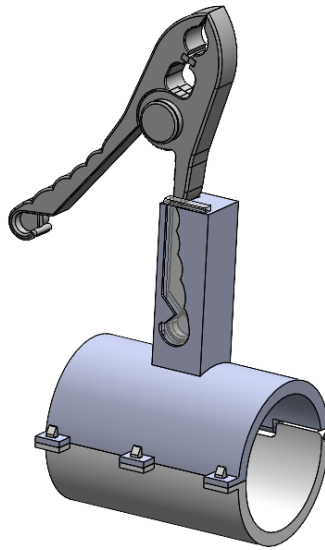


Figure 5.8: Detachable pylons.

5.5 X-1 Vehicle Release Mechanism

The X-1 Vehicle Release Mechanism was designed to minimize weight and drag, while ensuring precise deployment and secure attachment. The attachment consists of a servo motor and specialized servo arm. The servo motor is mounted to the top of the CF boom, along the center-line of the plane to maintain a balanced center of mass. The servo motor actuates a 3D-printed servo arm, translating the rotation of the servo motor from above the CF boom to key hole on the X-1 test vehicle's fuselage. The key hole on the vehicle's fuselage consists of balsa plates that ensure the vehicle is secured until the servo arm is oriented in a specific way. In one orientation, the servo arm anchors the vehicle. Once the servo motor rotates, the servo arm loses contact with the key hole, allowing it to disconnect from the fuselage. This setup is shown in Fig. 5.9. A lip will be installed behind the locked position of the servo arm to mitigate rotation and backwards movement of the glider during flight.

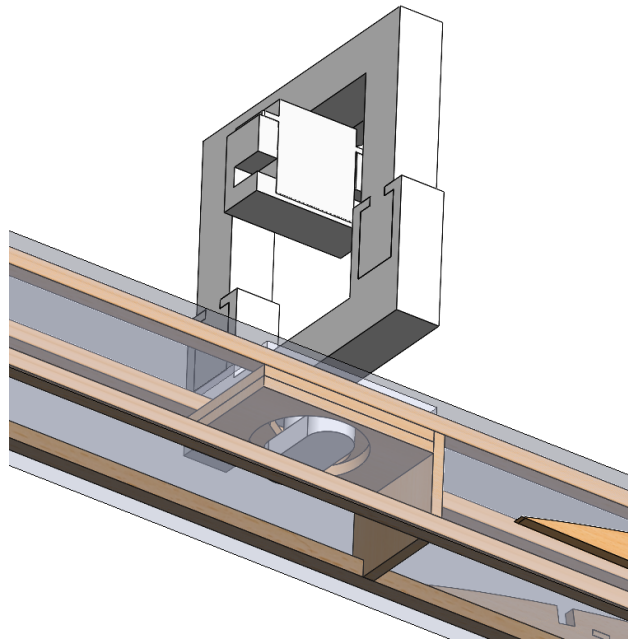


Figure 5.9: X-1 test vehicle release mechanism.

5.6 Weight and Mass Balance

Each mission has a different mass distribution. To estimate the CG of the vehicle, component masses and locations were determined from the CAD model and tabulated. The leading edge of the imaginary central wing rib was taken to be the origin, and a mutually orthogonal coordinate system was established with the positive X-axis pointing out the nose of the aircraft, the positive Z-axis extending below the aircraft, and the positive Y-axis directed out of the right-wing when viewing the aircraft from behind. The masses and absolute coordinates of the aircraft's major components are listed in Table 5.2. The weights and CG measurements for M2 and M3 were calculated including the payloads for each mission, respectively. Note that components such as bolts, glue, covering, and wire mass are not included in the table.

Table 5.2: Weight and balance for each mission.

Component	Weight [kg]	CG _x [mm]	CG _y [mm]	CG _z [mm]
Wing	1.323	-155.82	0	3.8
Fuselage	1.409	-228.94	0	98.38
VT	0.008	-1267.06	0.06	60.21
HT	0.050	-1220.82	0	139.4
Rear Landing Gear	0.161	-203.37	0	251.87
Front Landing Gear	0.008	172.5	-0.47	223.23
7x Control Servos	0.159	-556.15	-7.90	47.86
Control Electronics	0.258	-160.93	0	110.97
Propulsion Battery	0.438	8.89	0	103.76
Motor	0.451	222.91	0	76.2
M1				
Overall	4.451	-141.69	-2.56	84.45
M2				
4x Pylons	0.946	-78.68	0.4	145.76
4x Fuel Tanks	2.500	-60.3	-0.19	188.74
X1-Vehicle	0.200	-251.11	-2.01	186.41
Overall	8.097	-114.33	-1.73	120.48
M3				
4x Pylons	0.946	-78.68	0.4	145.76
4x Fuel Tanks	0.120	-60.3	-0.19	188.74
X1-Vehicle	0.200	-251.11	-2.01	186.41
Overall	5.717	-137.26	-2.38	91.51

5.7 Structural Performance

In order to verify structural performance of the aircraft, WUDBF utilized Solidworks FEA [16] to simulate the major structural components of the aircraft under their maximum expected load conditions.

5.7.1 Motor Forces

The motor forces are a significant source of stress on the aircraft, especially at static conditions such as takeoff. Based on Table 4.7, the static thrust was determined and then applied to the motor mount and fuselage. The results are shown in Fig. 5.10.

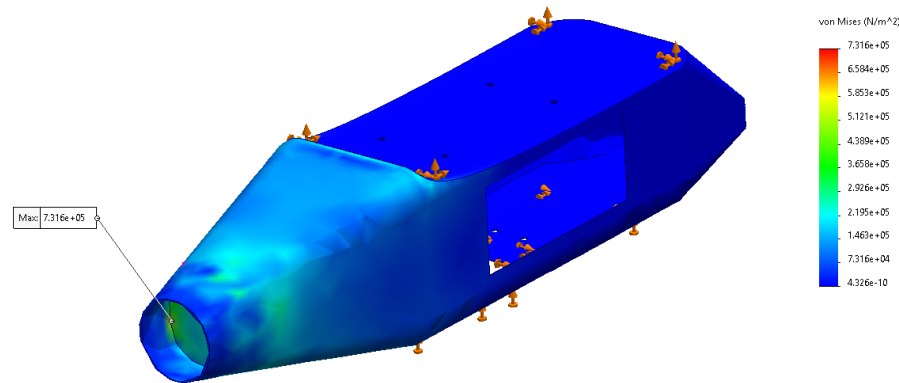


Figure 5.10: Von Mises stress plot on the fuselage.

The maximum stress was found to be 0.716 MPa where the motor mount meets the fuselage. This is well within the yield limits of both fiberglass and plywood [17].

5.7.2 Wing Load on Fuselage

Another significant load on the fuselage comes from the lift and weight forces during a turn. Using the load factor derived from Table 4.8 and the heaviest weight from Table 5.2, the maximum load was calculated and applied to the upper surface of the fuselage, where the wing would be attached. The results are shown in Fig. 5.11.

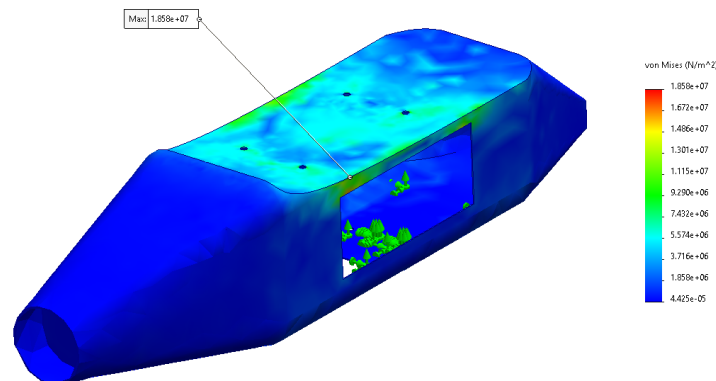


Figure 5.11: Von Mises stress plot on the fuselage.

The maximum stress was found to be 18.58 MPa, located at the edges of the top fuselage surface. Both the motor and the wing load stress are well below the ultimate stress of 521 MPa for E-Glass fiber, validating the design [17].

5.7.3 Wing Spar FEA

FEA was performed in SolidWorks on the CF spar to ensure the wing could withstand a wingtip loading test. The finite lift distribution was approximated by XFLR5 to be near elliptical. In order to input the distribution into SolidWorks, curve fitting was performed in MATLAB to generate a continuous distribution as shown in Fig. 5.12. Note that the lift distribution is plotted as a fraction of total lift as a function of unit span.

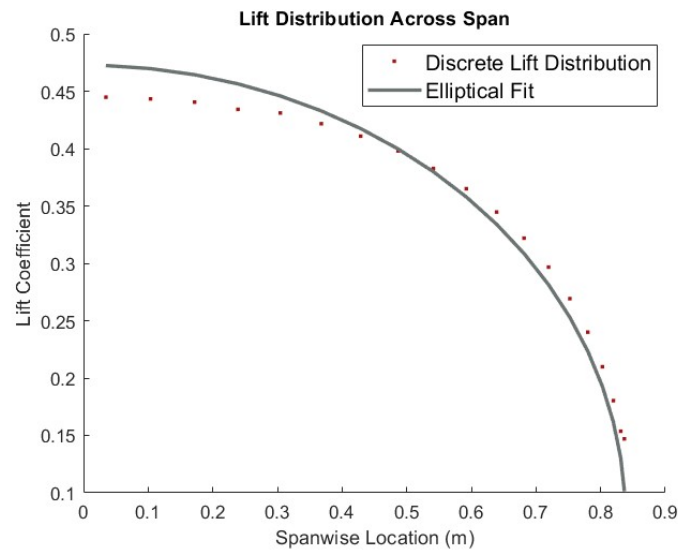


Figure 5.12: Finite and continuous spanwise function.

Using the fitted function and a 2.5 load factor, the stress and displacement of the wing was generated. The displacement field is shown in Fig. 5.13.

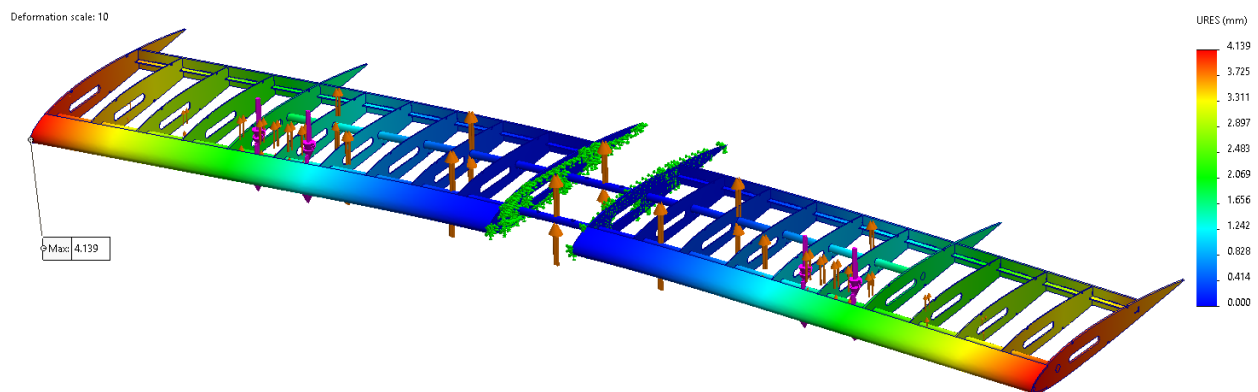


Figure 5.13: Displacement field of wing assembly under load.

The maximum displacement is 4.14 mm at the wingtips. The maximum von Mises stress was found to be 40.46 MPa, well below the 400 MPa yield stress of intermediate modulus carbon fiber [18].

5.8 Predicted Final Aircraft Performance

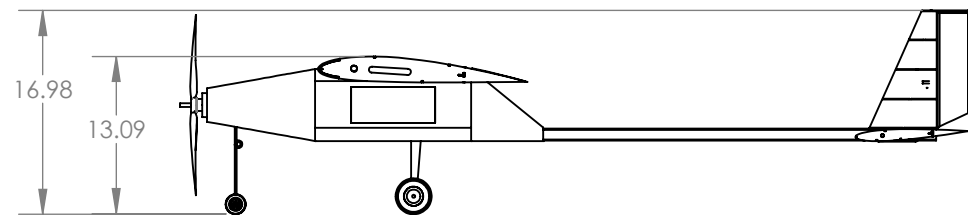
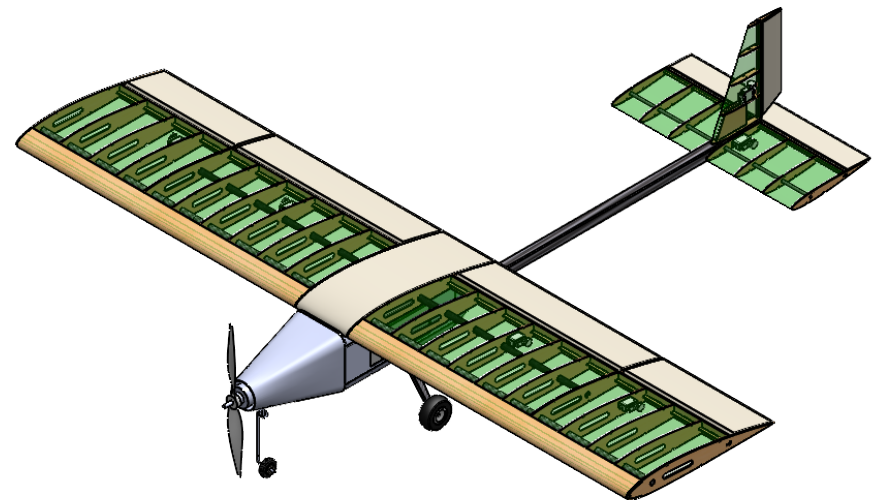
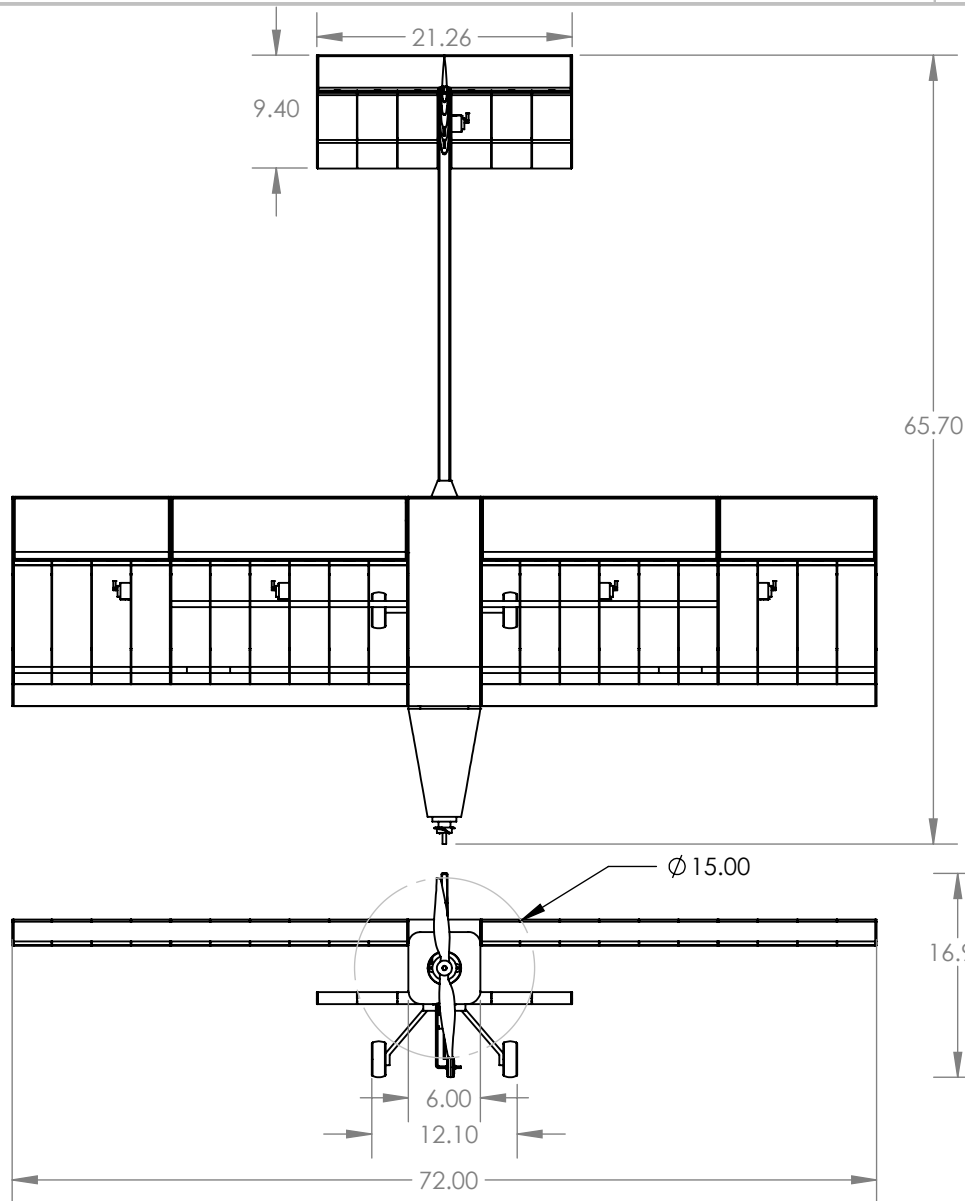
Using the updated weights in Table 5.2, Km_sim was utilized to simulate the expected flight and mission performance of the final aircraft. The extracted performance estimates are shown in Table 5.3. Normalized scores were calculated based on the maximum competition scores predicted as part of the analysis in Section 3.3.1. Note the M2 wing loading is slightly higher than the original target set by the constraint sizing. This change was deemed allowable since the T/W provided by the motor, 1.18, is higher than the original target of 0.950.

Table 5.3: Predicted final aircraft performance.

	Mission 1	Mission 2	Mission 3
Inputs			
All-Up Weight	4.451 kg	8.097 kg	5.717 kg
X-1 Test Vehicle Weight	-	0.20 kg	0.20 kg
Fuel Tank Weight	-	2.50 kg	0.12 kg
Wing Loading	6.01 kg/m ²	10.94 kg/m ²	7.73 kg/m ²
$C_{L,max}$	2.42	2.42	2.42
Parasitic Drag	0.081	0.081	0.081
Cruise Throttle	0.8	1	0.8
Outputs			
Takeoff Distance	7.90 m	12.7 m	10.8 m
$C_{L,cruise}$	0.350	0.350	0.350
$C_{D,cruise}$	0.094	0.094	0.094
Cruise Velocity	28.04 m/s	34.15 m/s	27.96 m/s
Number of Laps	3	3	5 (Before X-1 launch)
Mission Time at End of Final Lap	2.26 min	1.92 min	4.61 min (At end of lap after X-1 launch)
Scoring			
Normalized Score	1	2	2.08 (0 bonus), 2.16 (1 bonus), 2.28 (2.5 bonus)

B

B



A

All dimensions are in inches

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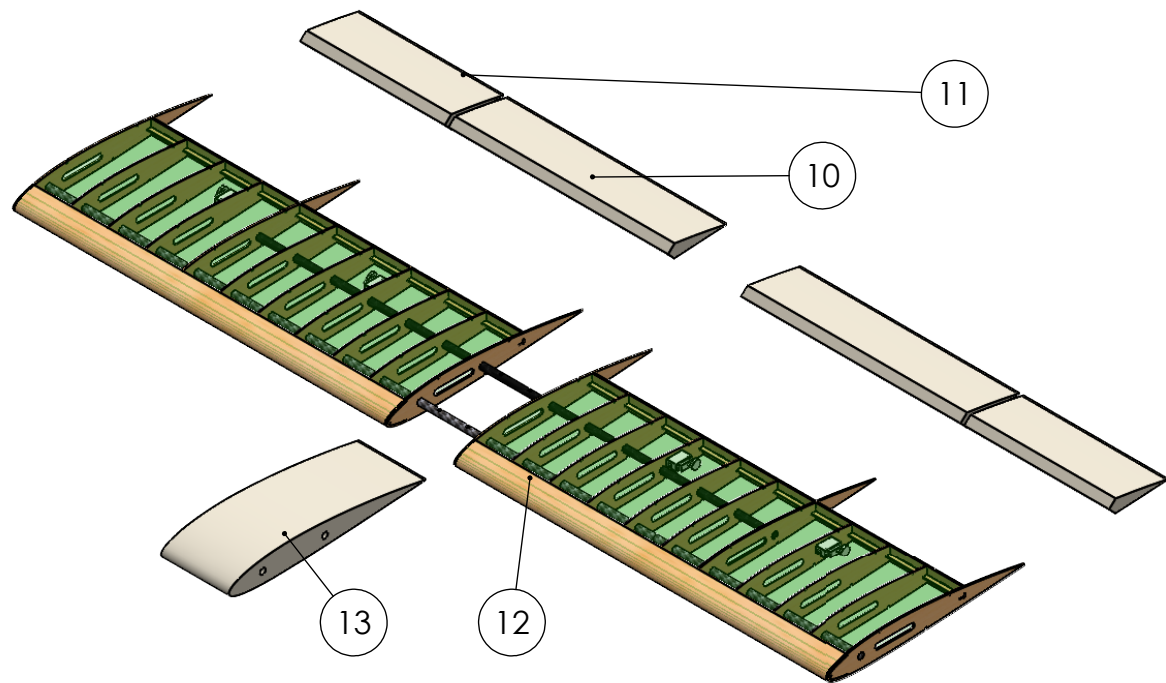
Aircraft Detail 3-View

SCALE: 1:16

2

1

B

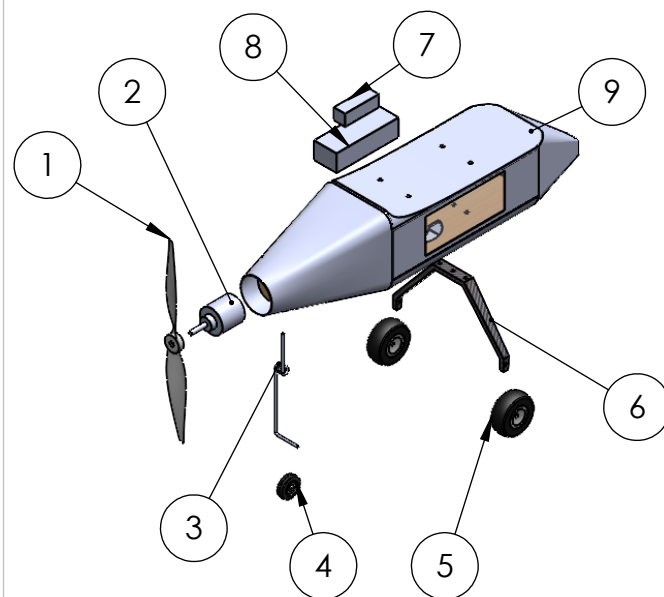


A

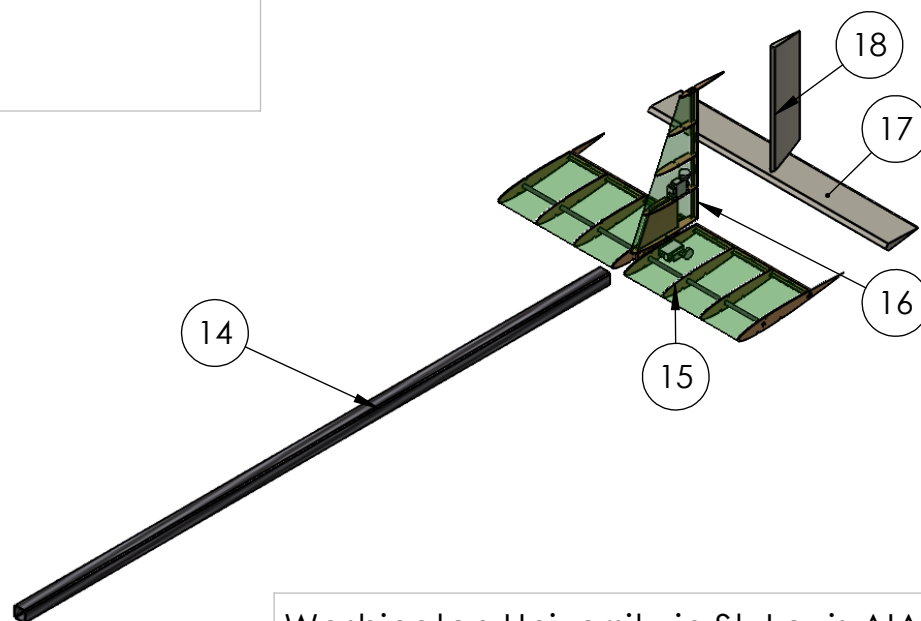
ITEM NO.	PART NUMBER	QTY.	Material
1	Propellor	1	Plastic
2	Motor	1	Aluminum
3	Front Landing Gear Spring	1	Steel
4	Front Landing Gear Wheel	1	Foam
5	Rear Landing Gear Wheels	2	Nylon
6	Rear Landing Gear	1	Carbon Fiber
7	Receiver Pack	1	-
8	Battery	1	Lithium Polymer
9	Fuselage	1	Fiberglass
10	Flaps	2	XPS Foam
11	Ailerons	2	XPS Foam
12	Wing	1	Balsa Wood
13	Wing Mount Box	1	XPS Foam
14	Carbon Fiber Boom	1	Carbon Fiber
15	Horizontal Stabilizer	1	Balsa Wood
16	Vertical Stabilizer	1	Balsa Wood
17	Elevator	1	XPS Foam
18	Rudder	1	XPS Foam

2

1



B



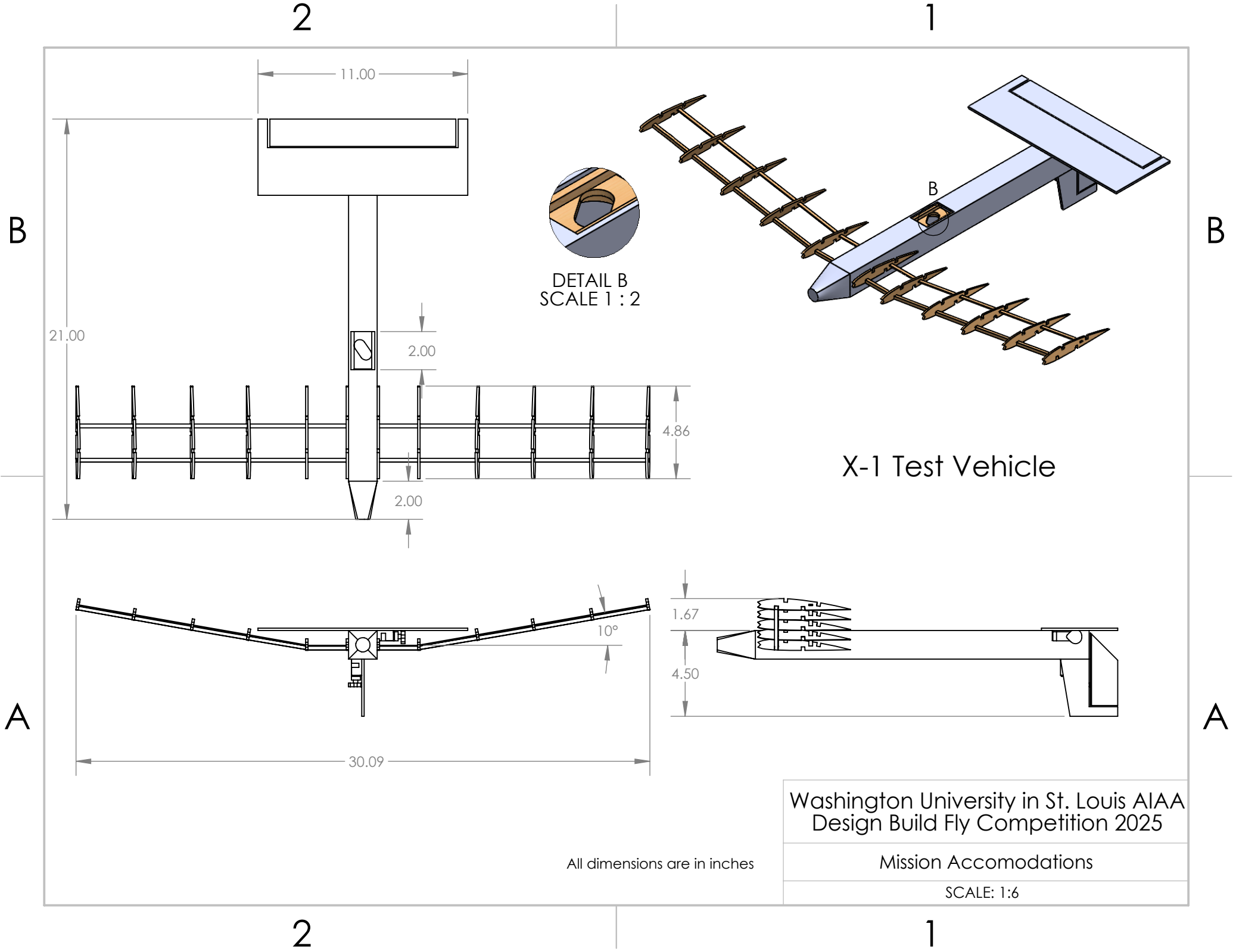
A

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Aircraft Exploded View

SCALE: 1:12

1

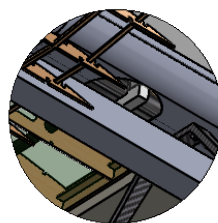
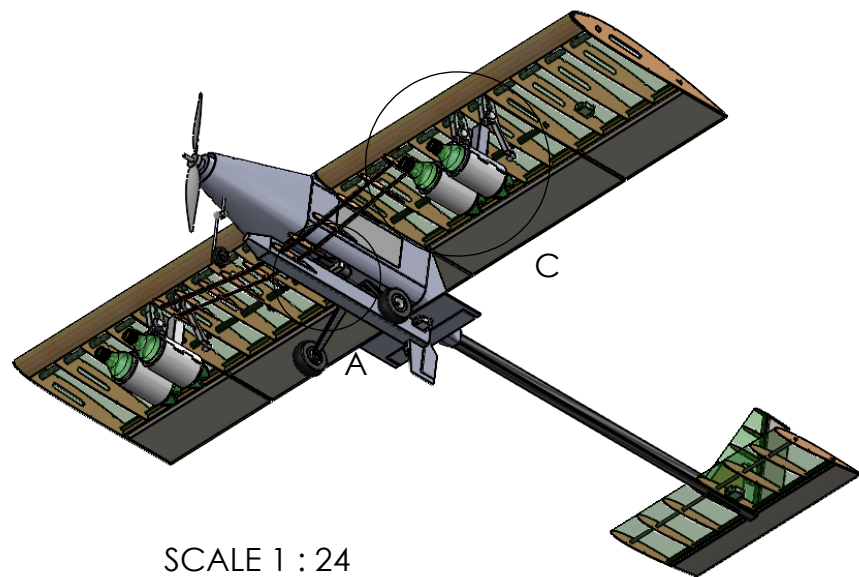


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Mission Accomodations
SCALE: 1:6

2

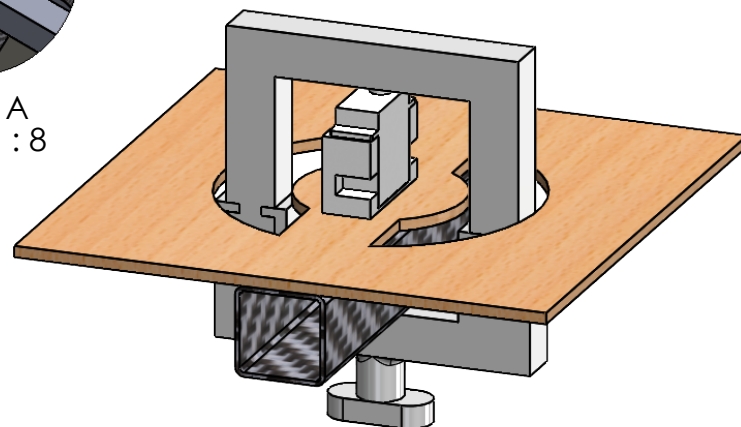
1

B



DETAIL A
SCALE 1 : 8

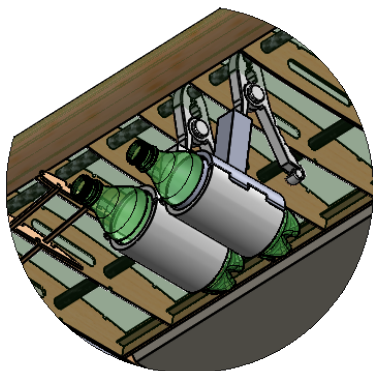
SCALE 1 : 2



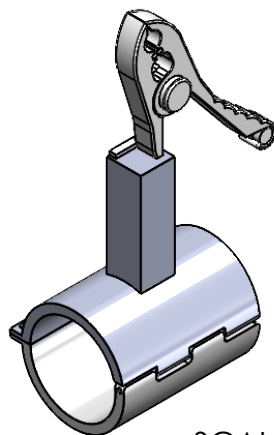
X-1 Test Vehicle
Release Mechanism

B

A

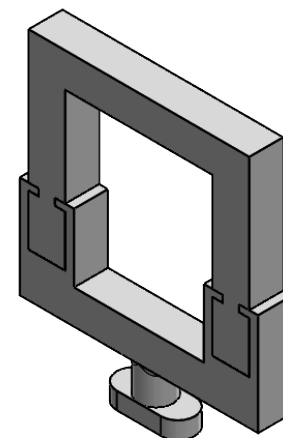


DETAIL C
SCALE 1 : 8



SCALE 1 : 4

Pylons



A

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Mission Accommodations

2

1

6 Manufacturing Plan

6.1 Selection of Manufacturing Methods

To construct the aircraft and various subsystems, multiple manufacturing methods were considered.

6.1.1 Selection Criteria

Methods were evaluated using key criteria such as ease of construction, efficiency, feasibility, and weight. Preference of materials was given to those which had been used in previous aircrafts, as manufacturing methods had already been refined. Additionally, cost was a factor given the limited budget of WUDBF.

6.1.2 Solid Foam

Due to the absence of internal bracing structures, solid foam components can be designed and fabricated quite quickly. They also weigh very little. Accurate foam shapes were achieved using a hot-wire cutter. Each component was layered with fiberglass to obtain sufficient strength to be used on the aircraft.

6.1.3 Additive Manufacturing

When considering ease of fabricating custom structures or components, as well as the streamlined workflow from CAD to printer, 3D printing has many advantages. However, 3D printed parts take long amounts of time to complete. Additionally, 3D printed parts are both heavier and weaker than those made from other materials, especially in the shear stresses between the layers and along the seems. WUDBF ultimately did not use any 3D printed parts for the main aircraft structure, however, smaller components such as servo mounts were 3D printed due to the cheap cost and ease of printing.

6.1.4 Composite Layup

Both carbon fiber and fiberglass layup methods were considered due to their high strength and low weight. Due to successful past experiences and previously refined manufacturing methods, fiberglass was chosen over CF layups. However, to reinforce major load bearing components on the aircraft, CF rods were used in addition to fiberglass layups.

6.1.5 Wood Construction

WUDBF is most experienced with wood construction manufacturing techniques. A laser cutter is used to rapidly cut out wooden components. WUDBF primarily uses balsa, basswood, and birch ply. Each of these woods offers different material properties. These properties are heavily considered when deciding where extra strength is needed or where weight can be saved. Assembly of wooden pieces is done using CA glue and two-part epoxy. CA glue is quick-drying and can form a strong bond between components and is the primary adhesive in wood construction. Epoxy offers a stronger alternative that is used for components that interact with heavier loads. The wooden frame is covered in Monokote, a plastic film that can be heated to adhere to the wood. The plastic film is pulled tight across the frame, as it shrinks when heated. This maintains the aerodynamic shape of the aircraft. Monokote is very light and relatively thin, making it susceptible to tearing; however, any damage can be easily repaired.

6.1.6 Process Selected For Major Components

A combination of solid foam, composite layup, and wood construction with Monokote coating were chosen as the primary construction methods for each of the main components. Solid foam and fiberglass composite layups were chosen for components, including the fuselage, due to the low weight and high strength of fiberglass. Wood construction was chosen for components, such as the wings and tail, due to its structural qualities as well as WUDBF's previous experience. Carbon fiber parts were added as reinforcement in key areas for structural support. In particular, the landing gear, tail boom, and wing spars are CF parts purchased online to include in

the aircraft.

6.2 Detailed Manufacturing Processes

The manufacturing processes used to construct the aircraft are presented below. Note that all details are primarily based on construction of the prototype aircraft, and minor adjustments may be needed for the final aircraft, which has not been built yet.

6.2.1 Wing and Tail

The wing and tail are primarily constructed using wood construction techniques. The wing consists of wooden ribs and stringers, as well as a CF spar spans the entire wingspan. Each wing is constructed separately and is designed to be easily removable. To ensure proper alignment, the ribs are placed in slotted basswood jigs. The CF rod and wood stringers are then inserted and secured using epoxy. A quarter-inch square stringer is attached to the leading edge and sanded down to achieve the desired curvature. Next, the mounting plates for the control surfaces are installed, the jigs are removed and sanded, and the wing is flipped. The leading edge is formed by soaking balsa sheets in warm water and folding them over the leading edge stringer. Once dry, 3D printed servo mounts are glued to the ribs, servos are installed, and the entire wing is covered in Monokote. The control surfaces are made using foam and fiberglass layup techniques. A hot-wire foam cutter is used to shape control surfaces, which are then covered in fiberglass and epoxied for reinforcement. Finally, hinges are attached to the control surfaces and then to the mounting plates on the wing to secure them in place. The tail is structured very similarly to the wings, using wooden ribs, stringers, and thin CF rods. Like the wing assembly, jigs are used to align the ribs, and a sanded stringer at the tip of the airfoil serves as the leading edge. The elevator and rudder are constructed separately and attached to the horizontal and vertical stabilizers, respectively, using hinges and epoxy. A central square CF rod extends outward from the fuselage, acting as the tail boom. Perpendicular holes are drilled into the rod to accommodate the horizontal tail spar, using a custom jig to ensure proper alignment. The tail is then epoxied to the square CF rod, servo mounts and servos are installed, and the entire assembly is covered in Monokote for a clean and aerodynamic finish.

6.2.2 Fuselage

The fuselage is constructed using a combination of solid foam and composite layup techniques. To shape the fuselage, a foam plug is cut using a hot-wire foam cutter. The plug is then covered with strips of fiberglass and saturated with epoxy. To maximize the strength-to-weight ratio, the fuse incorporates a total of four layers of fiberglass, with the strips oriented at specific angles: 0 degrees, 45 degrees, -45 degrees, and 90 degrees, with 0 degrees aligned to the width of the fuselage. Once the fiberglass layers are epoxied onto the foam plug, the entire assembly is placed into a vacuum bag to compress the layers and dry. Once dry, the fuselage is removed from the vacuum bag, and the foam is dissolved from the inside of the hardened epoxy shell using acetone. Openings are then cut to create the door and main electronics panel, and the CF tail boom is glued to the floor. Finally, custom laser cut birch plywood firewalls are installed.

6.2.3 Landing Gear

The landing gear is made of prefabricated CF and is secured to the main fuselage using nylon bolts. The bolts pass through the fiberglass fuselage and are secured to a wooden plate glued to the floor of the fuse for added reinforcement. A tricycle setup was chosen to allow freedom for steering, and to allow space for the glider.

6.2.4 Pylons

The pylons are constructed from commercially available clamps/fasteners in combination with a 3D printed fuel tank carrier.

Primary test objectives are listed in the table below by category. These objectives were achieved through several ground and flight tests described in Section 7.2.

Table 7.1: Breakdown of test objectives.

Category	Objective
Propulsion	Collect data from static thrust tests to confirm propulsion performance on the ground. Validate the propulsion system in the air during flight tests to confirm takeoff capability, cruise velocity, and endurance.
Manufacturing	Investigate manufacturing processes, in particular the benefits of composite fuselage construction.
Structures	Conduct wingtip loading tests at maximum AUW to verify structural integrity. Perform torsion and flutter tests to ensure that deflections are not excessive. Confirm the integrity of the landing gear via ground tests.
Ground Mission	Ensure rapid loading and unloading of payloads can be achieved.
Performance	Record flight data for analysis to confirm performance requirements are met. Obtain feedback from the pilot regarding handling qualities.

7.2 Ground Testing and Flight Testing

7.2.1 Propulsion Test

Static thrust testing was performed on the chosen propulsion system using an RCbenchmark Series 1585 Dynamometer mounted to a thrust stand. Using RCbenchmark software, thrust, RPM, current draw, and electrical power measurements were collected at incremental throttle levels. The data were then used to validate predicted performance and confirm the propulsion system met the necessary requirements.

7.2.2 Fiberglass Test

Fiberglass tests were performed to determine the ideal number and orientation of plies. Small strips of fiberglass prepared, layered, and epoxied into thin rectangular samples based on various test configurations. The test samples were then set up as simply supported beams with force applied to the center of the span. Each rectangle was tested until failure.

7.2.3 Ground Mission Test

Ground mission testing was conducted with WUDBF's chosen ground crew member and pilot, as well as a third participant to time the mission. The ground mission, as described in Section 3.1.4, was performed under testing conditions. The test was performed multiple times, until a consistent and quick time was achieved.

7.2.4 Wingtip Test

To validate the structural integrity of the fuselage, a wingtip loading test was performed at maximum takeoff weight. This test simulated a load case of 2.5 while verifying the aircraft could withstand the wingtip test required at competition. The results were then compared to the FEA performed in Section 5.7 to further validate the aircraft design.

7.2.5 Glider Test

In order to test glider capabilities WUDBF plans to release the test vehicle off of university buildings with permission from campus authorities. There will be a designated landing zone and scoring will be carried out similarly to competition to determine success of glider performance.

7.2.6 Flight Testing

A test flight was conducted to verify that the aircraft met design goals. A Pixhawk flight computer was used to capture telemetry during the test flight. Any data collected was then analyzed to extract measures of aircraft performance including speed, endurance, and lap times. Feedback from the pilot was also used to evaluate the stability and flight characteristics of the aircraft. Conclusions from flight testing were used to inform changes in

the design going from the prototype to the final aircraft. For full details of the testing schedule see Table 7.4

7.2.7 Ground Inspection and Pre-Flight Checklists

Table 7.2: Ground inspection checklist.

Component	Tasks
Motor	☐ Motor secured ☐ Propeller free of damage
Fuselage	☐ No cracks or tears ☐ Electronics are connected and secured
Wing	☐ No cracks or tears ☐ Bolted to fuselage ☐ Wingtip test
Payload	☐ Payload is positioned correctly and secure
Control Surfaces	☐ No cracks or tears ☐ Servos connected to receiver ☐ Control surfaces can deflect fully
Empennage	☐ No cracks or tears ☐ Control surfaces move without interference
Landing Gear	☐ Front wheel secured and servo connected ☐ Rear wheels secured

Table 7.3: Pre- and post-flight checklists.

Flight Checklists
Pre-Flight
☐ Payload secured ☐ Propulsion battery level check ☐ Receiver battery level check ☐ CG position confirmed ☐ Wingtip test ☐ Receiver turned on ☐ Control surfaces test ☐ Landing gear steering test ☐ Range check ☐ Motor run-up ☐ Hatches and payloads are secure
Post-Flight
☐ Throttle down ☐ Arming plug removed ☐ Propulsion battery disconnected ☐ Receiver battery disconnected

7.3 Testing Schedule

Table 7.4: Testing schedule.

Date	Type	Category	Objective
Prototype			
10/19/2024	Ground	Performance	Glider configuration and weight testing.
1/19/2025	Ground	Manufacturing	Fiberglass strength testing.
2/15/2025	Ground	Propulsion	Static thrust test to confirm propulsion performance.
2/15/2025	Ground	Structures	Wingtip test to confirm structure.
2/20/2025	Flight	Performance	Shakedown flight. Test prototype design.
2/20/2025	Flight	Performance	Attempted M1 simulation flight. Confirm TOFL + flight. (CRASHED).
Competition Aircraft (Planned)			
3/9/2025	Ground	Performance	Ground mission test.
3/30/2025	Flight	Performance	M1 Simulated flight. Evaluate performance.
3/30/2025	Flight	Performance	M2 Simulated flight. Evaluate performance.
3/30/2025	Flight	Performance	M3 Simulated flight. Evaluate performance.
4/6/2025	Flight	Performance	Practice for competition.

8 Performance Results

8.1 Ground Test Results

8.1.1 Propulsion Test Results

Results from the static thrust test confirmed that the propulsion system generally met the desired performance requirements. Test data compared to eCalc predicted performance in the table.

Table 8.1: Predicted vs. Actual Performance

Parameter	Prediction	Stall Test Result	Flight Test Result
RPM	10571	9791	10143
Thrust [kgf]	6.435	7.239	–
Max Current [A]	106.1	91.1	109.0
Electrical Power [W]	2495.9	2259	2720

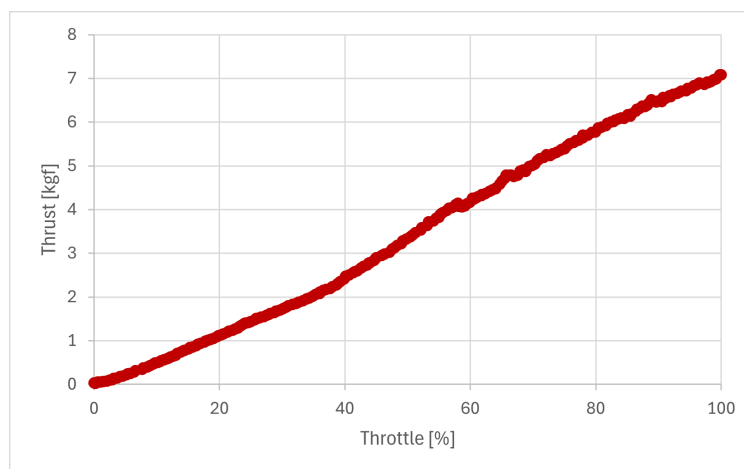


Figure 8.1: Static Thrust Test Results

Based on the static testing, the motor drew slightly more burst current than predicted. This is not an issue, however, since the ESC was sized with significant margin. In addition, the static thrust appears greater than what was expected due to eCalc's predicted 10% +/- error. Later flight testing confirmed that the propulsion system delivered the desired takeoff capability and cruise performance

8.1.2 Fiberglass Test

Fiberglass testing revealed that symmetric laminate configurations were significantly stronger than those non-symmetric ones. Additionally, the tests confirmed that WUDBF's desired strength-to-weight ratio could be achieved with a total of four plies. Thus, WUDBF decided to construct the fuselage using four layers of fiberglass strips, oriented at 0 degrees, 45 degrees, -45 degrees, and 90 degrees, where 0 degrees aligned with the width of the fuselage.

8.1.3 Ground Mission Test

The numerous simulated ground mission tests allowed WUDBF's chosen ground crew member to achieve a consistent timed score. Furthermore, the ground crew member determined the optimal positioning for themselves as well as each component to ensure the greatest effectiveness and speed.

8.1.4 Wingtip Test

The aircraft was successfully supported by just the wingtips. The measured deflection was extremely slight, around 2 mm in each wing. This agrees with the FEA testing done on the wing spar in Section 5.7 (predicted total deflection of 2.07 mm).

8.2 Flight Testing

The range sensor connected to the Pixhawk that allowed communication between the receiver and transmitter was disconnected shortly after take-off. Immediately after the disconnect, elevator control was lost, and the aircraft entered fail-safe mode and was unrecoverable. No flight test data was captured due to the crash. The wing structure remained intact, however had numerous rips in the Monokote coating. The fuselage was also largely undamaged, which verified its strength. This failure motivated a number of changes in the electronics integration, including adding extra reinforcements to electronics inside the main fuselage.

8.2.1 Feb. 20th, 2025 - Flight (Prototype)

Ursa Major performed one flight during the Feb. 20th test flight of the first prototype. The preliminary design can be seen in Fig. 8.2. Table 8.2 summarizes the outcomes of the Feb. 20th test flight.



Figure 8.2: Ursa Major on Shakedown Flight of Feb. 20 Test Flight.

Table 8.2: Test flight outcomes.

Flight Description	Result
Shakedown	The pilot began trimming the aircraft and was able to achieve stable flight characteristics. After the aircraft was in steady flight for 25 seconds, connection with the receiver was lost and the airplane entered fail safe mode. This has prompted more careful placement and alignment of the receiver antenna. Additionally, the wings seemed loose during flight. Thus, more reinforcement will be added to the fuselage connection.

References

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